NATIONAL GEOGRAPHIC
HISTORY
PRIESTESSES OF AMUN
DIVINE WORSHIP IN THE NEW KINGDOM
COMBAT IN THE COLOSSEUM
ROME'S GREAT GLADIATORS
PLUS:
The Mystery of Dorothy Arnold
New York's Oldest Cold Case
THE SCOTTISH POMPEII
THE WELL-PRESERVED VILLAGE OF SKARA BRAE

NATIONAL GEOGRAPHIC
HISTORY
VOL. 12 NO. 5

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At Roman gladiator schools, enslaved people, prisoners of war, criminals, and debt-ridden volunteers made up the vast majority of recruits. Here they would receive room and board and specialized training in the art of combat according to their fighting profile. The stout and sturdy might become helmeted *secutores*, while the light-footed became trident-wielding *retiarii*. A string of victories could secure the favor of the masses—and a single defeat was often enough to stir the crowds into demanding a swift execution.

The Gladiator Mosaic, found on the Borghese estate at Torrenova, Italy, in 1834.

SCALA, FLORENCE

24 A Unique Neolithic Settlement

Off the north coast of Scotland lie the Orkney Islands, where a cluster of houses stood preserved for more than four millennia, until a violent storm in 1850 left them exposed. Since then, a series of surveys have revealed glimpses into the life of a prehistoric island community. From the rough stone box beds to the fine bone tattooing implements found there, Skara Brae is an archaeological treasure trove.

40 The Powerful Priestesses of Egypt

During the New Kingdom, Egypt was at the height of its splendor. At this time, the great Temple of Amun within the Karnak Temple Complex hosted priestesses who were masters of dance and song, responsible for delighting Amun with their performances. At their head was a figure called the God's Wife, a privileged woman whose power would eventually rival the pharaoh's.

74 Michelangelo's Neglected Masterpiece

The tomb of Pope Julius II (a wealthy artistic patron) was envisioned by Michelangelo as a colossal monument that would sit under the dome of St. Peter's Basilica in Rome. However, after years of negligence by the renowned sculptor, the project would eventually be demoted to a mere cenotaph, and the affair has been dubbed by art historians as "the tragedy of the tomb."

92 A Banquet for a Phrygian King

A large mound rises from the plains of central Türkiye that for millennia has harbored the remains of a great Phrygian ruler—long suspected to be King Midas, with the "golden touch." When archaeologists reached the tomb, in 1957, they found no gold but unearthed a magnificent burial hoard and clues to the true identity of the deceased.

Omphalus drinking bowl, found in the tomb.

MET/SCALA, FLORENCE

The Gundorf Stater is on display in the State Museum of Archaeology, Chemnitz, Germany, until February 2027.
© ARCHAEOLOGICAL HERITAGE OFFICE OF SAXONY GUNDORF STATER AVERS/REVERS - A. KALTOFEN

CELTIC GOLD COIN

Gold at the End of the Rainbow

A detectorist in Leipzig, Germany, has discovered a rare Celtic coin.

A small but extraordinary discovery may have rewritten Saxony's numismatic history. In a field northwest of the German city of Leipzig, Daniel Fest, an amateur metal detectorist, unearthed a 2,200-year-old quarter stater made of almost pure gold in July 2025.

Until recently, only two Celtic coins were known to exist in Saxony, and one, discovered in the 19th century, has since been lost. Although Saxony was located beyond the Celtic settlement areas, this discovery proves the existence of trade.

"The coin wouldn't have been in circulation but was probably a status object or part of a symbolic deposit belonging to the local elite who traded with Celtic communities … The find not only enriches Saxony's archaeological record but also underlines the importance of collaboration between citizens and scientists in protecting cultural heritage," explained archaeologist Regina Smolnik.

THE OBVERSE OF THE COIN shows the head of an animal, possibly a deer, while the reverse features an open necklace, a star, and a sphere: common motifs among the Celtic communities of northern Bohemia. Due to its convex shape, the coin belongs to a type called *Regenbogenschüsselchen* (rainbow cups). This term comes from the folk legend that claims that treasures can be found where a rainbow touches the ground. Coins of this type used to appear, glinting in the sun, in recently plowed fields—after rainfall washed away the soil.

Coins in the Brandenburg State Archaeological Museum
GETTY IMAGES

Sacred Stone of the Southwest is on the Brink of Extinction

Centuries ago, Persians, Tibetans and Mayans considered turquoise a gemstone of the heavens, believing the striking blue stones were sacred pieces of sky. Today, the rarest and most valuable turquoise is found in the American Southwest— but the future of the blue beauty is unclear.

On a recent trip to Tucson, we spoke with fourth generation turquoise traders who explained that less than five percent of turquoise mined worldwide can be set into jewelry and only about twenty mines in the Southwest supply gem-quality turquoise. Once a thriving industry, many Southwest mines have run dry and are now closed.

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Dorothy Arnold: New York's Oldest Cold Case?

The mysterious disappearance of the heiress, more than a century ago, remains a subject of fascination. Was she abducted or killed—or did she run away?

Dorothy Arnold's disappearance has enthralled and confounded ever since December 12, 1910. On that icy Monday morning, the 25-year-old socialite left her Upper East Side mansion to go dress shopping for her little sister Marjorie's upcoming debutante ball. She told a friend that she was shopping with her mother, and told her mother that she would rather shop alone.

As the second-born child of the wealthy and well-connected Arnold family—whose roots trace back to the Mayflower—and niece to a Supreme Court justice, Dorothy casually carried $30 in cash (almost $1,000 in today's money) and was decked out in the height of New York fashion. She left her other clothes, jewelry, and her passport at home. By all accounts, Dorothy was in good spirits. She stopped at Park & Tilford for a half-pound box of chocolates, and then bought a book of humorous epigrams (titled *Engaged Girl Sketches)* at Brentano's—charging both to her father's accounts. Just before 2 p.m., at the corner of 27th Street and Fifth Avenue, Dorothy bumped into a friend, sparking a short but lively chat, where she mentioned plans to stroll home through Central Park. Then, she waved goodbye and vanished forever.

The oldest cold case of a missing person in New York City, the sensational disappearance of Dorothy Harriet Camille Arnold has mystified detectives and citizens ever since.

Without a Trace

The Arnold family knew something was amiss when Dorothy didn't come home for dinner. When she hadn't returned by nightfall, they called around to friends. And when she was still missing in the morning, the Arnolds hired a friend, the lawyer John Keith, to look for her. When his search of hospitals, morgues, and jails was unsuccessful, they decided to enlist a team of private investigators from the Pinkerton Agency. They didn't call the police, however, for a total of six weeks.

Although the above may sound suspicious, this was a logical reaction for a high-society family. "I think the attitude about the police was that they were for drunks, immigrants, and criminals," says Jane Vollmer, Dorothy's great-niece. Her younger brother, Mark Vollmer, concurs: "We take for granted now that, of course, you'd call the police," he says, "but people were very private back then."

Perhaps worse than the police was the media, whose rampant speculations could be (and ultimately were) socially devastating. But when the Pinkerton detectives came up empty, patriarch Francis Rose Arnold reluctantly called the New York Police Department and held a press conference on January 25. He described Dorothy in detail: five foot four, 140 pounds, pretty, and wearing an ankle-length navy dress and a velvet hat atop her "full pompadour." He offered a hefty reward of $1,000 (about $33,000

FINAL KNOWN LOCATION

THE LAST PLACE anyone saw socialite Dorothy Arnold was at the corner of Fifth Avenue and 27th Street in New York City. It was here that she ran into a friend, Gladys King, before disappearing into the busy street, turning back two times to wave goodbye. Allegedly, the friends stopped briefly to chat, and discussed Marjorie's upcoming ball, before Gladys left to meet her mother for lunch. Dorothy told her she was going to walk home through Central Park.

The intersection where Dorothy was last seen, in a photo from 1893. ALAMY

Dorothy Harriet Camille Arnold, in a photo taken not long before her disappearance.
ALBUM/ALAMY

today) for any information that led to her whereabouts.

The next day, the heiress's disappearance landed on the front page of the *New York Times*. Though her father was "prostrated with grief and worry" and her mother "on the verge of nervous collapse," the article noted that police believed Dorothy had eloped, and did not suspect foul play or suicide; however, that theory was ultimately ruled out. "Miss Arnold was not of a secretive nature," the *New York Times* wrote. "Although she has been admired by many young men, a representative of the family said last night that she had never been engaged, as far as they knew... There is no trace of insanity in the family, and the young woman had never shown signs of a troubled mind, although she was devoted to books and spoke several languages."

A Secret Affair

On February 15, the *Daily News* dropped a bombshell: The September before she went missing, Dorothy had pawned family jewelry to fund a weeklong tryst at Boston's Hotel Lenox.

"The young woman had never shown signs of a troubled mind, although she was devoted to books and spoke several languages." —*New York Times*

An Aspiring Writer With a Double Life

DOROTHY ARNOLD wasn't like most girls of her era and status. A recent literature and language graduate of Bryn Mawr, a liberal college in Philadelphia, Dorothy aspired to more than a well-positioned marriage; she wanted to become a writer and had been (unsuccessfully) submitting short stories to *McClure's Magazine*, including one titled "The Poinsettia Flames." She had asked her father for a studio apartment in bohemian Greenwich Village, a move that was denied to her, and when her family mocked her writerly efforts, she used her ample allowance for a private post office box, permitting her to receive correspondence away from prying eyes. The charred remains of what might have been a rejected manuscript were later found in her fireplace, and pamphlets advertising transatlantic steamship voyages were found lying in her room.

PARK & TILFORD CANDIES, where Dorothy stopped to buy chocolates, at the corner of Fifth Avenue and 59th Street. The retailer was known for its high-quality confectionery and art deco tin boxes. According to the shop's clerk, she appeared to be in good spirits when she made her purchase at 1:45 p.m., the day she disappeared. ALAMY

Though she'd told her family she was visiting a friend in Cambridge, in reality, Dorothy was with George Griscom, Jr., an unemployed engineer in his 40s with whom she had been having clandestine liaisons since her days at college. Friends said Griscom had proposed marriage to Dorothy, but her parents staunchly disapproved and had forbidden her from seeing him.

The Arnolds denied it all, but it was later revealed that Dorothy's mother and older brother, John, had sailed to Europe weeks earlier to find and confront her boyfriend. ("Griscom licked by young Arnold in foreign hotel," read one headline). Griscom denied any knowledge of Dorothy's disappearance, vowing both his love for her and his intentions to marry her upon her return.

"Obviously, the path laid out for Dorothy was to marry a 'suitable' man from an equally well-heeled, well-connected family and bring up children to do the same," says historian Silvia Pettem, who included Dorothy Arnold in her book, *Cold Case Chronicles: Mysteries, Murders & The Missing*. Dorothy's aspirations to move out, write, and date whomever she wanted prove that the heiress was already deviating from that narrow path. "Dorothy just didn't fit the mold that was expected of her," says Pettem.

> The Arnold family angered the police by their unwillingness to collaborate in complete honesty.

Diagram of Dorothy published in *The Evening World* on January 26, 1911, describing what she wore on the day of her disappearance.
LIBRARY OF CONGRESS

Theories and Claims

Despite her sunny disposition and privileged lot in life, a surfaced letter to Griscom revealed a melancholy side to Dorothy: "*McClure's* has turned me down," she wrote. "Failure stares me in the face. All I see ahead is a long road with no turning." And then, foreboadingly, "Mother will always think an accident has happened." In fact, it appears that her mother never gave up hope of Dorothy's return, while Keith,

THE UNLIKELY LOVER

GEORGE GRISCOM, JR., who liked to be addressed as Junior, was a bachelor still living with his wealthy parents. Among his many improprieties, Griscom had a previous engagement to another rich heiress, who had called off the wedding the night before. Dorothy's father told the press he "would have been glad to see her associate more with young men than she did, especially some . . . of brains and position" and that he didn't approve of men "who have nothing to do." On his return to the U.S., the press reported that Griscom was "locked in" by his father, for fear that he might say something that would upset the Arnolds.

MEMBERS OF THE GRISCOM FAMILY ON THE BOARDWALK AT ATLANTIC CITY.

MR. AND MRS. GEORGE S. GRISCOM, SR.

SAY AKED MAY GO TO SAN FRANCISCO PULPIT

Church of 49ers Gives Fifth Avenue Baptist Pastor Warm Reception.

PREACHED THERE SUNDAY

Report That He Will Accept Call Held in Trustworthy Quar-

GEORGE S. GRISCOM, JR.
Who said that he expected to marry the missing Dorothy Arnold.

George Griscom, Jr., and his parents snapped by the press in Atlantic City.
LIBRARY OF CONGRESS

the lawyer and family friend, was convinced she died by suicide.

In an era before social security numbers, disappearing was as easy as changing your name and hairstyle. Certainly, plenty of pretty brunettes looked close enough. In 1914, an attorney claimed that a woman living in Los Angeles under the name "Ella Nevins" claimed she was the missing socialite. "If you don't believe I am Dorothy Arnold," the *Gettysburg Times* quoted, "ask my sister, Marjorie. She will know why my father doesn't answer my letters." There were countless similar claims, however, as well as fake ransom notes, so-called "sightings," and postcards from "Dorothy" that inundated the Arnold family. One, apparently in Dorothy's handwriting, simply said, "I am safe."

Other theories abounded: In April 1914, Bellevue House, a "maternity hospital" in a Pittsburgh suburb known as the "House of Mystery" was raided by police. They arrested C.C. Meredith, a physician offering illegal abortions, who allegedly implied Dorothy had been cremated in the basement furnaces, supposedly after a botched operation. Some think Dorothy became pregnant during her week in Boston and had traveled to Pittsburgh on the advice of Griscom for a procedure that ultimately killed her. Francis Arnold called this theory "ridiculous."

Two years later, a convict confessed he'd been paid $250 by a rich man who matched Griscom's description to move and bury the heiress's body in the cellar of a suburban New York residence. Police excavated homes in the area but found nothing. The man, said Arnold, spewed "utter nonsense."

While Pettem's *Cold Case Chronicles* ultimately leans toward the latter as Dorothy's most likely fate, the author and historian has since changed her mind. "I used to think she died in a botched abortion," says Pettem, "but now I think she walked away and started a new life." Publicly, Dorothy's family said they believed she was kidnapped and murdered, a theory her father maintained until his death in 1922. Privately, he spent a million dollars in his search for Dorothy. It's possible Francis Arnold couldn't find his daughter because she didn't want to be found; it's also possible he knew everything but buried the truth to avoid a scandal. "Dorothy's papers and letters went to my mother, who destroyed them all," says Vollmer. "If there was a family secret, my mother might have never known. And if she did, it went to the grave with her."

— Rosemary Counter

THE NAVAL BATTLE in the harbor of Syracuse. Engraving from Cassell's *Illustrated Universal History,* 1882. Colorized.

PHOTOAISA

The Sicilian Expedition, Athens's Big Mistake

Thucydides recounted the catastrophic Athenian campaign against the Sicilian city of Syracuse in dazzling style in a work of great objectivity for the fifth century B.C.

In his *History of the Peloponnesian War,* the Greek historian Thucydides describes the Athenian expedition to Sicily of 415-413 B.C. as "by far the most costly and splendid Hellenic force that had ever been sent out by a single city up to that time." This expedition would also be Athens's worst disaster: a tactical error that would have fatal consequences.

Six years earlier, in 421 B.C., Athenian general Nicias had achieved a truce in the war that Athens and Sparta had been waging. But many in Athens yearned to resume hostilities. Among them was a young aristocrat called Alcibiades, who was, according to Thucydides, motivated only by wealth and fame. Thucydides recounts that in early 415 B.C., an embassy from the Sicilian city of Segesta (or Egesta in ancient Greek) arrived in Athens. Invited to speak before the citizens, the Segestian ambassadors explained that their city was at war with its neighbors Selinus and Syracuse. In an attempt to garner military backing, the Segestians warned that if their city were defeated, Syracuse would conquer Sicily in its entirety, and proceed to ally itself with Sparta. They also assured the Athenians that they had the support of other Sicilian cities, and that Segesta would finance the war—both were grave exaggerations. Nicias was unconvinced, and thought

A FORCE TO BE RECKONED WITH

THUCYDIDES EXPLAINS THAT in Corcyra (now Corfu), the Athenians divided their army into three, with commanders Alcibiades, Nicias, and Lamachus each taking charge of a division. They then sailed to Sicily via the Gulf of Taranto (between the "heel" and "toe" of southern Italy). The military fleet was accompanied by an additional 30 ships carrying provisions, as well as "bakers, stone-masons, and carpenters, and the tools for raising fortifications." Many other vessels "followed the armament voluntarily for purposes of trade."

NGM MAPS

it would be foolish to intervene. Alcibiades, however, persuaded the Athenian public that Sicily could prove to be a lucrative source of income.

In city squares and gymnasiums across Athens, young and old alike began to speak enthusiastically of the potential expedition. Despite Syracuse being some 500 nautical miles away, the citizens supported the idea, convinced that they had a powerful navy and ample financial reserves. The Athenian *Ekklesia* (assembly) voted to send a fleet to Syracuse, led by Alcibiades, Nicias, and Lamachus.

Then, one morning, amid preparations for the fleet, Athens awoke to find all the city's *hermae* (stone pillars bearing the face, torso, and sometimes genitals of the god Hermes) mutilated. Suspicion fell on Alcibiades and his friends. Their arrogance and excess marked them as people of low morals. Thucydides recounts that some Athenians accused them of not only committing this sacrilege but also parodying the secret religious rites known as Eleusinian Mysteries during a night of drunken revelry. Despite the scandal, the Athenians decided that Alcibiades should set sail for Syracuse. He would be tried on his return, if they didn't recall him sooner.

Alcibiades' fellow citizens considered him to be arrogant and immoral.

Bust of Alcibiades. Roman copy of a fourth-century B.C. original. Capitoline Museums, Rome

AKG/ALBUM

Off to War

On the day of departure 5,100 hoplites (heavily armed infantry) and about 1,300 light infantry soldiers (including archers and slingers), along with Athenian allies and noncombatant personnel, embarked at dawn. Thucydides tells how all the Athenians went down to the port to bid farewell to their friends and loved ones. A herald led a prayer, a hymn was sung, and libations were offered before the ships weighed anchor.

However, upon arriving in Sicily, nothing was as they had expected: Segesta did not have the resources

SYRACUSE was the most prosperous Greek city in Sicily. Its theater, built in the fifth century B.C., was completely rebuilt in the third century B.C. and again in Roman times.
ADOBE STOCK

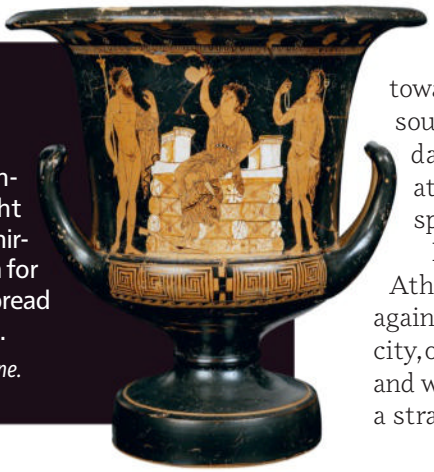
needed to wage a war, nor were the other cities on the island prepared to make an alliance with Athens.

Nicias proposed a brief show of force in support of Segesta before returning to Athens. But Alcibiades and Lamachus favored remaining in Sicily to assault Syracuse. So the Athenians moved their army to Catania, north of Syracuse, which became their center of operations. At Catana, an Athenian herald arrived on the *Salaminia,* a sacred trireme, with orders for Alcibiades to return to Athens to stand trial. Alcibiades appeared to have complied and immediately set sail. But after he anchored at Thurii in southern Italy, Alcibiades escaped to Sparta, where he sought asylum. Plutarch writes that Alcibiades, upon learning that the Athenians had sentenced him to death, declared, "I'll show them that I'm alive." In an act of defiance, he encouraged the Spartans to send a contingent to Sicily to aid the Syracusans. Meanwhile, Nicias and Lamachus attacked Hyccara, a village in the north of the island, and enslaved its inhabitants to raise money. They then advanced by sea from Catana toward Syracuse, landing on a beach south of the city's Great Harbor. At dawn, they launched a surprise attack, but failed and withdrew to spend the winter in Catana.

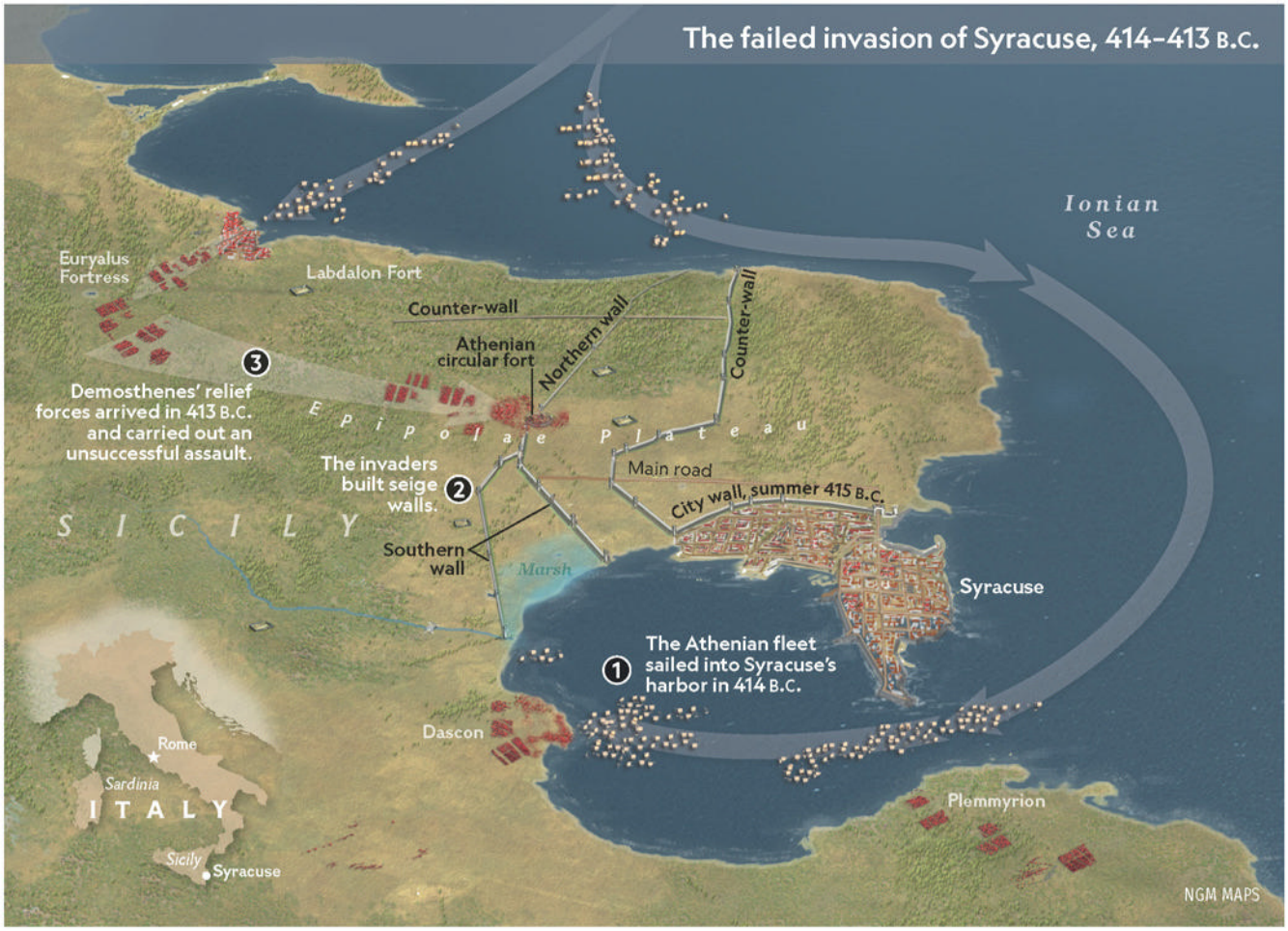
In the spring of 414 B.C., the Athenians landed in Syracuse once again. Nicias set about besieging the city, ordering the construction of forts and walls across the Epipolae plateau, a strategic high ground. Thucydides

LIFESAVING VERSES

SOME OF THE FEW Athenian prisoners who managed to return home visited the playwright Euripides to tell him that the Sicilians were admirers of his works. They explained that in return for singing Euripides' verses, they would receive bread and water, or even freedom from enslavement.



Sicilian krater with a scene from Euripides' tragedy *Alcmene.*
ERICH LESSING/ALBUM



recounts how the Syracusans built their own defensive walls, while they charged the Athenians. Lamachus was killed in one of these clashes, and Nicias appealed to Athens to send help.

The Arrival of Gylippus

The tide truly turned against the Athenians when Spartan general Gylippus arrived. The Athenians hadn't finished constructing their siege walls, and the Spartans were able to meet with the Syracusans. That summer, Gylippus razed the Athenian fortifications and then pushed the Athenians back to their base at Plemmyrion (south of the Bay of Syracuse).

The next year, Athenian reinforcements arrived. Demosthenes, their commander, attempted to recapture the Epipolae plateau in a night raid, but many Athenian hoplites, unfamiliar with the terrain, fell to their deaths in the darkness. Shortly before, Gylippus had taken Plemmyrion and seized provisions, supplies, and even trireme rigging. Then, the Syracusans blocked the exit from the bay, trapping what remained of the Athenian fleet inside, where they suffered a definitive defeat. The Athenians began to retreat on foot toward Catana with some 40,000 men, leaving behind the wounded. They were attacked along the way, and on the sixth day of the march, Gylippus cut off Demosthenes' retreat. Nicias was able to continue, but when his men reached the Assinarus River, the soldiers, wild with thirst, rushed forward in disorder. This is when Gylippus's men struck. As Thucydides recounts, the slaughter left many bodies "piled one upon another in the stream." Nicias surrendered.

Gylippus led the captured army to Syracuse, where Nicias and Demosthenes were executed. The Athenians and their allies were then imprisoned in the city's quarries. Thucydides describes in graphic detail how many of the prisoners, crammed into a deep and narrow gorge, died of disease, hunger, or thirst, and that their bodies were left rotting among the excrement, with the "intolerable stenches." According to Thucydides, never had Athenians suffered such a defeat: "Their fleet, their army, everything was destroyed, and few out of many returned home."

The Athenians, who remembered the splendor with which the formidable expedition had set out, lamented its pathetic end. This was an unmitigated disaster, which paved the way for the final defeat of Athens at the hands of Sparta less than 10 years later.

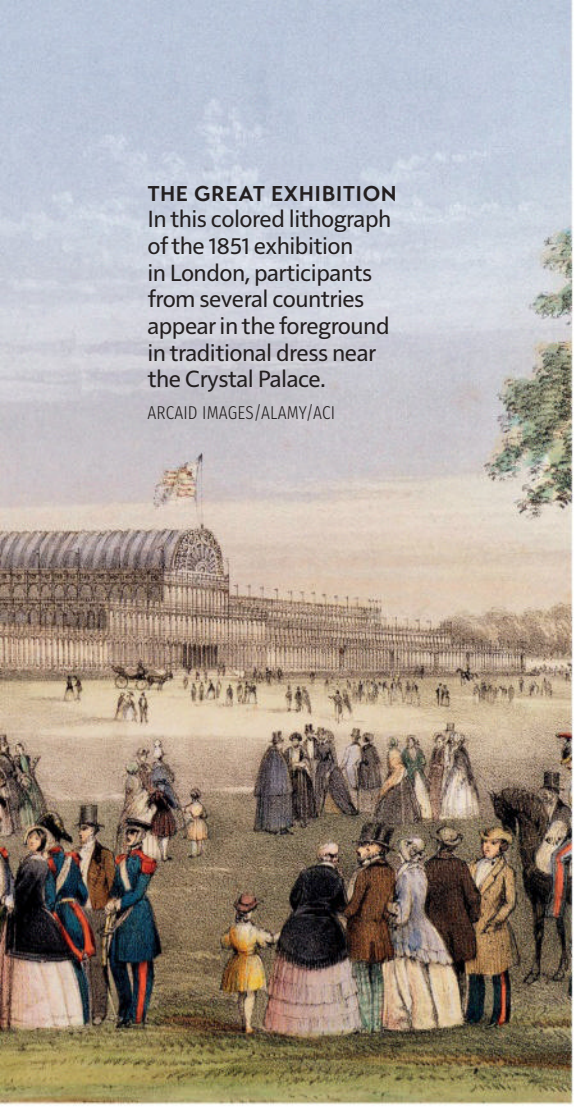
—Juan Pablo Sánchez

The Crystal Palace Captivates the World

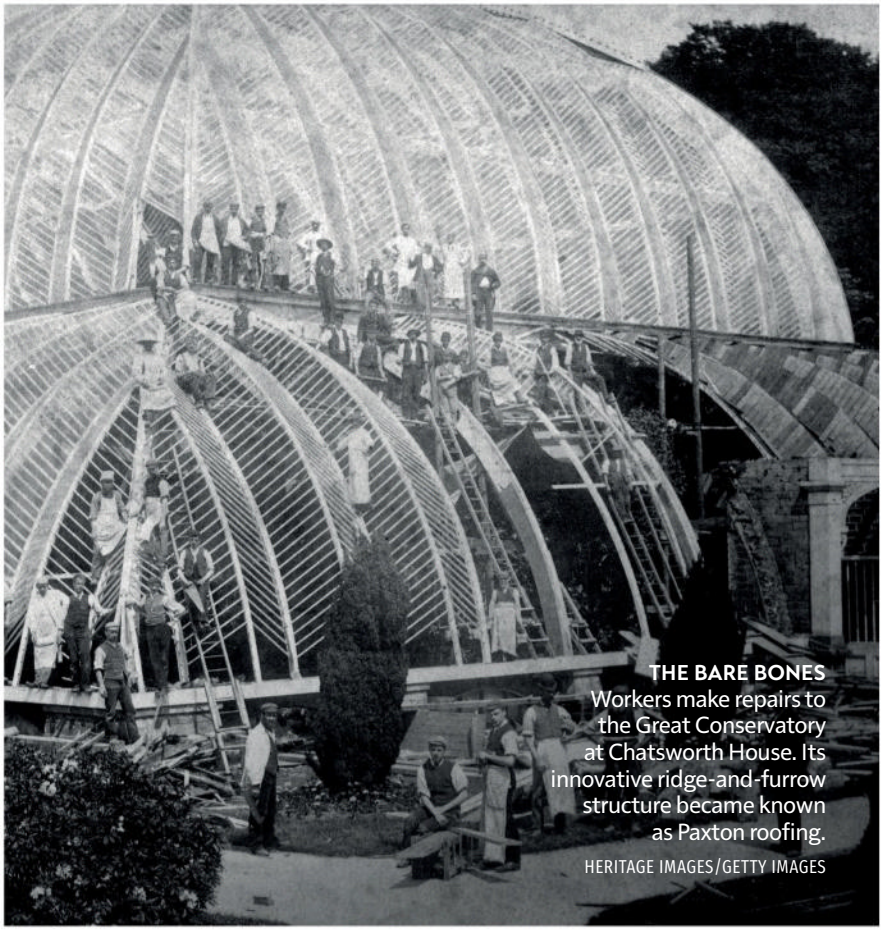
The first world's fair was held in England, inside a colossal building of iron and glass that pushed the boundaries of engineering and seemed to defy physics.

In 1851 London hosted the first ever world's fair, formally the Great Exhibition of the Works of Industry of All Nations. France had already mounted 11 major exhibitions to showcase its own national industries, the first of which was held in 1798. Britain was less experienced, with only three national exhibitions of industrial products since 1847. But several elements made Britain a superior candidate as host for a more ambitious undertaking. One was the national economy, which did not place restrictions on imported products. Also key were two individuals who drove the process forward: Prince Albert (husband of Queen Victoria) and Sir Henry Cole—a writer, designer, and artist who revolutionized the connection between Victorian art and industry. Together they fostered the idea of holding an international exhibition in London and set up a royal commission to make it a reality.

To house the exhibition, the plan was to build an impressive temporary structure in London's Hyde Park that would showcase British modernity and industrial might. In 1850 the exhibition's building committee (which included famous engineers Robert Stephenson and Isambard Kingdom Brunel) launched an open competition, receiving 245 designs. Two of these proposed a building made of iron and glass, an approach that caught the



THE GREAT EXHIBITION
In this colored lithograph of the 1851 exhibition in London, participants from several countries appear in the foreground in traditional dress near the Crystal Palace.
ARCAID IMAGES/ALAMY/ACI



THE BARE BONES
Workers make repairs to the Great Conservatory at Chatsworth House. Its innovative ridge-and-furrow structure became known as Paxton roofing.
HERITAGE IMAGES/GETTY IMAGES

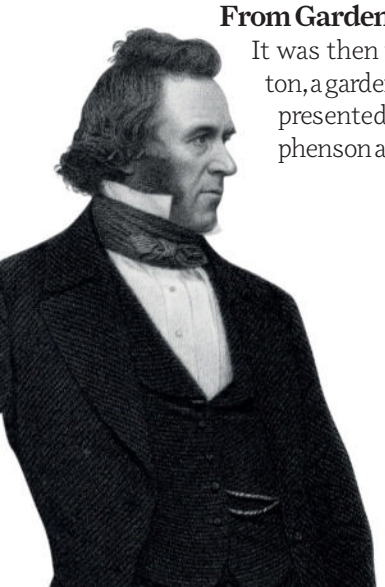
committee's attention. Yet neither of these designs (submitted by Frenchman Hector Horeau and Irishman Richard Turner and his son Thomas) were deemed suitable because they were too laborious, expensive, or permanent. The competition was declared void, and the committee's own design proposal was also rejected.

From Gardener to Architect

It was then that Joseph Paxton, a gardener by profession, presented a project to Stephenson and also published it in the *Illustrated London News*. Paxton's drawings made such an impression on the public that the committee decided to award him the commission to construct the exhibition headquarters. His design was given the moniker the Crystal Palace by a *Punch* humor magazine, and the name stuck.

Paxton wasn't an architect, but in the course of his work as head gardener at the stately Chatsworth House, he had gained experience in constructing large greenhouses for the cultivation of exotic plants. One notable creation was Chatsworth's Great Conservatory (1840), once the largest glass building in the world, whose ribbed structure was inspired by the water lily leaf. For the development of the exhibition building, Paxton enlisted the help of various engineers, including Charles Fox and William Barlow. The fact that neither Paxton, Fox, nor Barlow were architects meant they were open to the new possibilities of industry, guided by their experience in the construction of greenhouses (in the case of Paxton), railway engineering (Fox), and civil engineering (Barlow) rather than by architectural trends.

Paxton's design was radical. The Crystal Palace was to measure 1,851 feet long by 408 feet wide, with a maximum height of 108 feet, allowing several trees to remain within it. It was the largest building ever constructed in London at the time. Nevertheless, it was built at relatively low cost, aided by the repeal of Britain's glass tax.



Paxton drew on his experience in building greenhouses for his design of the Crystal Palace.

Sir Joseph Paxton (1801–1865) was a gardener, landscape designer, and politician.
ALBUM

Throughout most of history, the materials used in construction had been primarily stone, wood, adobe, and brick. However, the industrial revolution made iron widely available. From the early 19th century onward, iron became a key material in construction. Likewise, the introduction of new methods for glass production in 1832 by the Chance Brothers and Company made it possible to manufacture larger, stronger, and cheaper glass in sheets. It would be the Chance Brothers' glasswork that supplied the glazing for the Crystal Palace—almost a million square feet of it. The panes measured just over four feet long, a size that could only be manufactured at scale in Britain.

Module by Module

In order to handle such a large number of components, it was essential to keep to a limited number of standardized pieces that could be assembled into a series of identical prefabricated modules. Paxton and his collaborators designed a structural module measuring 24 feet by 24 feet, which basically consisted of four columns and four girders made from cast iron. This module could be added to the structure from above or to the side; it was a three-dimensional addition module. This modular approach was highly innovative, not only setting the Crystal Palace apart from already existing iron-and-glass greenhouses and winter gardens but also making it a landmark development in the history of architecture.

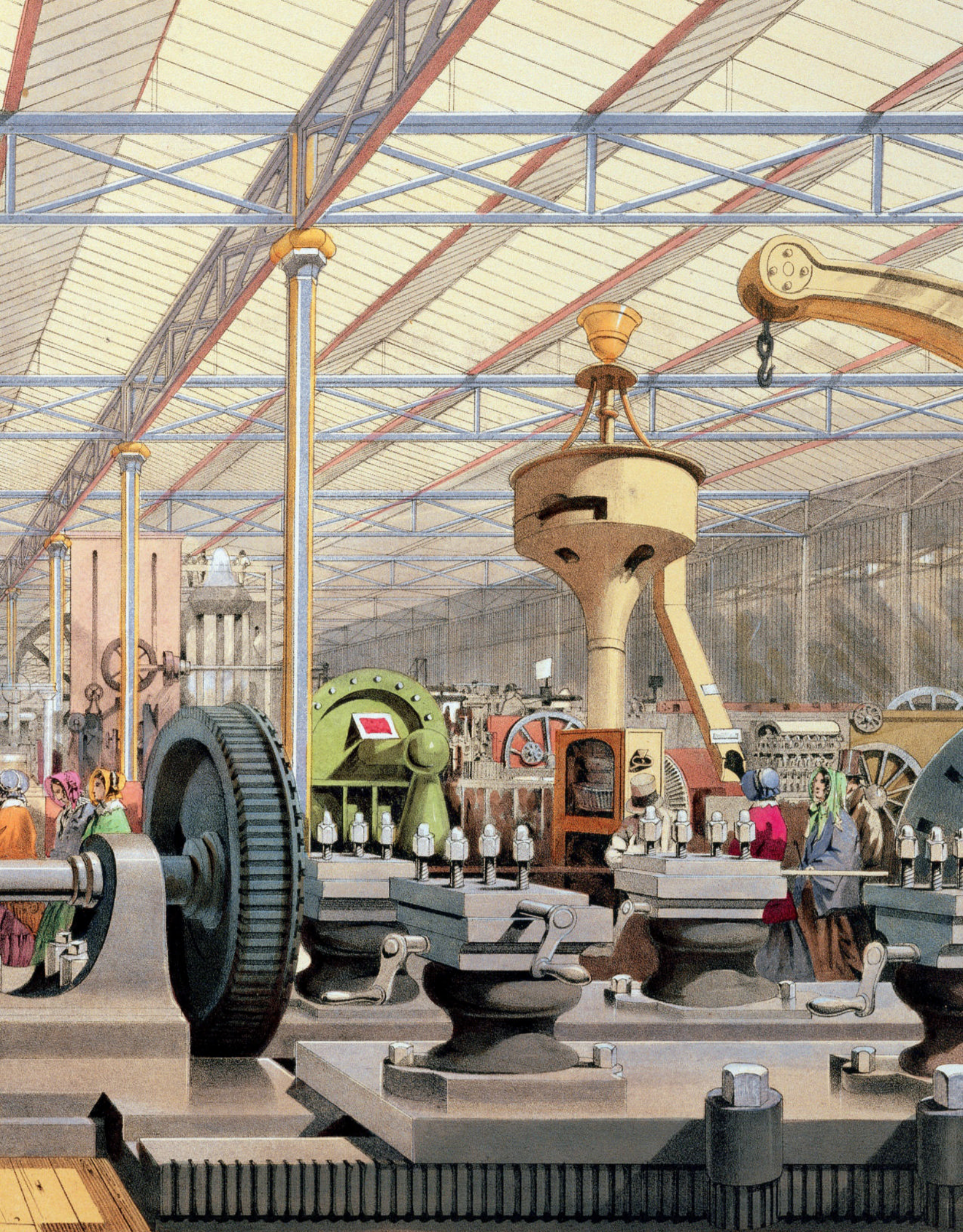
In 1851 the internal combustion engine was nascent; the main driving forces used were still human and animal power. The steam crane, which would later be used in 1889 to construct the Eiffel Tower, had just been patented by Sir William Fairbairn, so its use was not yet widespread. Therefore, the weight of any prefabricated component was limited to one ton in order to facilitate assembly.

The firm Fox, Henderson & Co. began construction in July 1850. The total assembly time did not exceed

A Spectacle of Countless Marvels

THE LONDON WORLD'S FAIR comprised over 100,000 objects arranged in four large categories dedicated to raw materials, machinery, manufacturing, and fine arts. Part of the machinery section was powered by a boiler room housed outside the building, allowing visitors to observe the entire process of cotton textile production in action, along with a huge hydraulic press and a printing machine reeling off the *Illustrated London News*. The other sections featured everything from the largest diamond known at the time (the Koh-i-Noor) to musical organs, photography, firearms (including Colt pistols), and Egyptian archaeological artifacts.

The moving machinery section of the 1851 exhibition
THE STAPLETON COLLECTION/BRIDGEMAN/ACI



A QUEEN'S CEREMONY Standing on the dais with Prince Albert, Queen Victoria opens the Great Exhibition at the Crystal Palace in Hyde Park.
GRANGER/ALBUM

six months, an extraordinarily short period, which caused both anxiety and fascination from the press and public.

When the building was almost finished, it was found that birds nesting in several trees now encased within the Crystal Palace had become trapped inside. Worried about bird droppings ruining the exhibits, and unable to use firearms for fear of breaking the glass, the organizers faced a conundrum. It was allegedly solved when Queen Victoria went to the Duke of Wellington for advice. His words: "Try sparrow hawks, ma'am."

The finished building faced further challenges. The structure—despite all its sophistication—remained essentially a greenhouse in nature, producing high temperatures. To counteract this, movable ventilation slats were added to the facade. Wooden paneling shut out the sun on the south facades, and awnings were stretched over the flat roofs. Nonetheless, maximum temperatures soared to 104°F.

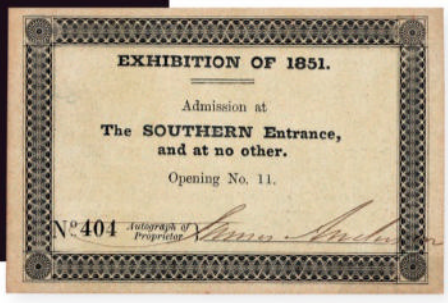
The heat caused the iron girders to expand to such an extent that reports from the time explain it was possible to see with the naked eye how some of the building's pillars were skewed out of their vertical alignment. This phenomenon had been anticipated, and some girders had been replaced by wooden beams, to little effect.

The Inauguration

Queen Victoria opened the Great Exhibition on

CLASS DIVISIONS

THE PRICE OF ADMISSION was initially one pound (approximately $165 today), but it eventually fell to as low as one shilling (about $8 today). Well-to-do visitors' anxiety about mixing with the masses was assuaged by "shilling days" being restricted to Monday through Thursday.


EXHIBITION OF 1851.
Admission at
The SOUTHERN Entrance,
and at no other.
Opening No. 11.
No 401 Autograph of Proprietor

Entrance ticket signed by the beneficiary
BRIDGEMAN/ACI

The Crystal Palace Goes up in Flames

ON THE NIGHT of November 30, 1936, Sir Henry Buckland, general manager of the Crystal Palace, was making rounds of the building, accompanied by his daughter Chrystal and their dog, when he saw a small fire had broken out. Despite the intervention of 89 fire trucks, 438 firefighters, and 749 police officers that same night, by dawn the building had been devastated by the flames. The loss of the building that had been an architectural symbol of the industrial revolution had a huge public impact. Winston Churchill described it as marking "the end of an age." This was a common sentiment. To many, the loss of the Crystal Palace symbolized the end of Victorian Britain.

Firefighters battled the blaze (left) but the site was ravaged by flames (right). Only water towers survived.

ALAMY/ACI (LEFT); ROGER VIOLLET/AURIMAGES (RIGHT)

May 1, 1851, to much fanfare. The regent described the event in her diaries as "a day to live for ever." She would return to the exhibition over 30 times. It is estimated that more than six million people—roughly equivalent to a third of the national population at the time—visited during the exhibition's six-month run.

Various contemporary accounts mention the exhilarating sense of space that the building inspired. Lothar Bucher, a German political refugee and publicist, described his impressions as follows: "We see a fine network of lines, but without any reference point to gain an idea of the distance from the eye or the actual size. The side walls are too far apart to be taken in with a single glance, and instead of resting on an opposite wall, the eye wanders along an endless perspective that disappears into the haze. We do not know whether the roof floats a hundred feet above us or a thousand, whether the ceiling is flat or formed by a multitude of small roofs, for there is no casting of shadows, which otherwise helps the soul to understand the impressions of the optic nerve."

Copycat Palaces

The Crystal Palace made a profound impression around the world. Suddenly, other countries wanted to have their own version. One was built in New York for the world's fair of 1853. The following year, the Glaspalast ("glass palace") was built in the Bavarian capital for the Munich industrial exhibition.

After London's Great Exhibition ended, the original Crystal Palace was dismantled and relocated to Sydenham Hill (South London) in 1854, albeit with certain modifications and additions that altered its shape and size. The building was used for celebrations, exhibitions, music festivals, circus performances, and other events until its unfortunate demise 82 years later. This area of the capital city remains known as Crystal Palace today.

The legacy of the 1851 exhibition is still alive today. The royal commission used its considerable profits to purchase land in London's South Kensington district, whose development under the initiative of Prince Albert gave rise to what is now known as Albertopolis. The area is dedicated to the arts, science, and learning, including the Natural History Museum, the Victoria and Albert Museum, and the grand concert venue the Royal Albert Hall.

—Isaac López César

The Dodo Bird, an Icon of Extinction

Composite dodo skeleton preserved in the Natural History Museum in London.
BRIDGEMAN/ACI

This flightless bird, endemic to the island of Mauritius, died out within a century of the arrival of Dutch explorers. It has since become a byword for its own demise.

Located in the Indian Ocean, around 700 miles east of Madagascar, the island of Mauritius was uninhabited until 1598, when a Dutch expedition arrived. The colonists named the island after the stadtholder Maurice, Prince of Orange. Further back, in the 10th century, Arab merchants had visited the island, and the Portuguese had been familiar with it since 1507. It was Portuguese sailors who discovered that the island was populated by extraordinary-looking birds that stood three feet tall and had long, hooked bills. Unable to fly, the bird could be caught easily, or clubbed to death. Its name may originate from the Portuguese *doudo* (stupid), although it is more likely to have derived from its Dutch nicknames, including *dodoor* (sluggard) or *dodaar* (from *dodaersen*, meaning "fat behind").

The island of Mauritius, created from a deep-sea volcanic eruption, was lush with endemic flora and fauna that had evolved separately from those on the African continent. With no natural predators, the dodo (*Raphus cucullatus*) appears to have evolved from a pigeon, with its closest living relative being the Nicobar pigeon. The dodo bird fed on snails and seeds, and its large beak also allowed it to swallow fruit. To survive the scarcity of food during the dry season, it built up reserves of fat during the wet season, and had adapted to living entirely on the ground.

Scene from *Alice's Adventures in Wonderland*, in which the dodo awards Alice her own thimble. 1865.
LOOK AND LEARN/BRIDGEMAN/ACI

The Arrival of the Dutch

The Dutch built Fort Frederik Hendrik on the eastern coast of the island to serve as a supply station for ships heading to the East Indies. Here, lemons, oranges, and sugar cane were cultivated using enslaved people. The colonists introduced dogs, cats, pigs, goats, cows, and monkeys. This first settlement lasted 12 years, and saw exploitation of the island's indigenous ebony forests. The forests sheltered the dodo, whose unflattering Dutch monikers included *walghvogel*, or "disgusting bird," since its meat was seen as far less appetizing than that of the island's plentiful turtledoves. (However, Vice Admiral Wybrand van Warwijck, in his 1598 diary, admitted that the dodo's stomach and breast were relatively tasty.)

Although the Dutch did not technically hunt the birds to extinction, the animals they brought with them finished the job. It was easy for these invaders to find and consume the dodos' eggs, as their nests were built on the ground and left unprotected. It

The (likely oversized) dodo of the Dutch painter Roelandt Savery, circa 1626.
NATURAL HISTORY MUSEUM, LONDON/BRIDGEMAN/ACI

is believed that the hapless species was extinct by 1690.

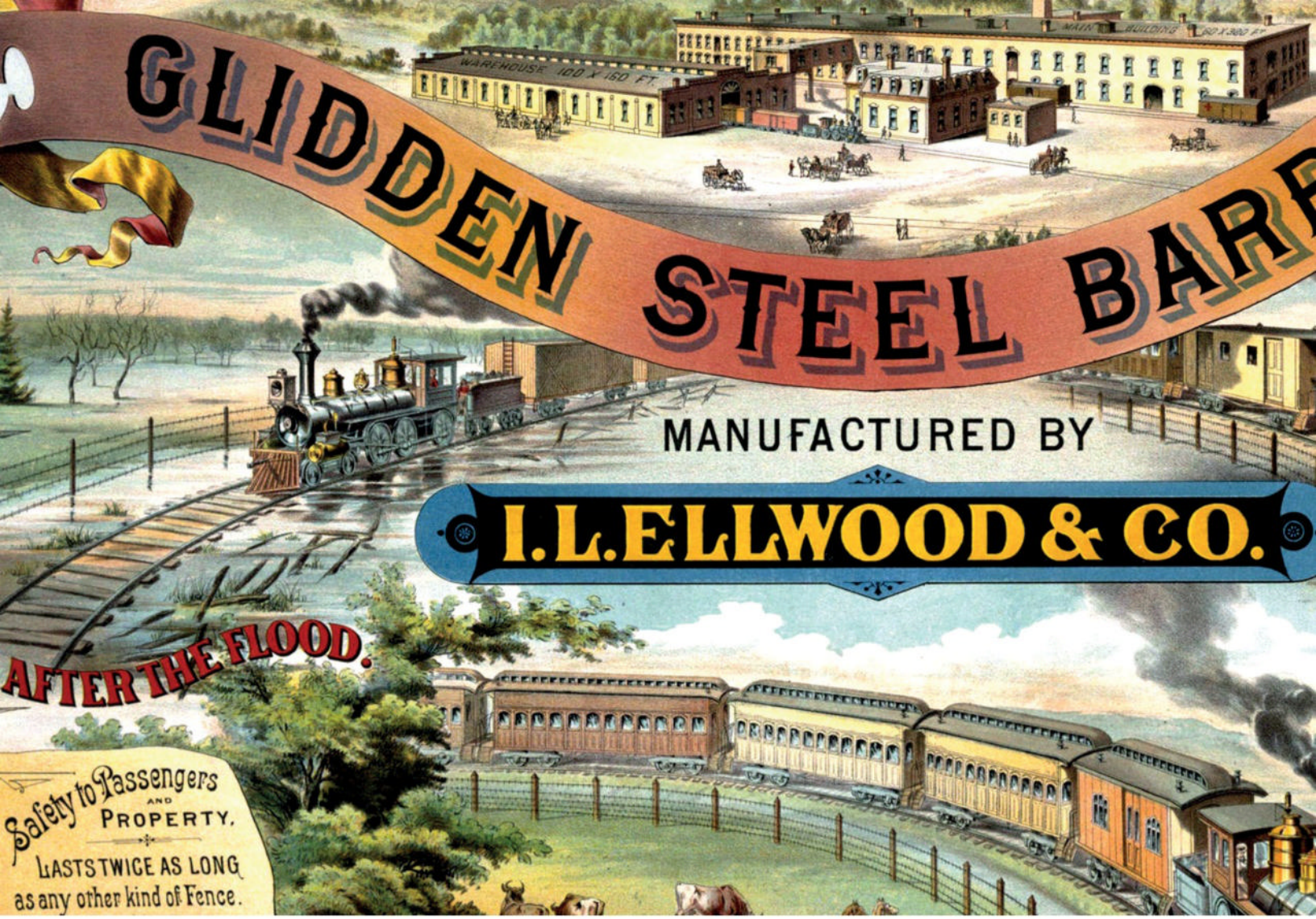
Gone But Not Forgotten

The dodo rose to fame posthumously, reappearing in early Victorian scientific writing and as a model exhibited at the 1851 Great Exhibition in London. But most prominent would be its character in Lewis Carroll's 1865 work *Alice's Adventures in Wonderland*. The bird's place in the popular imagination was reflected in the phrase "dead as a dodo" (likely first used in 1852 in *Household Words*, a literary magazine edited by Charles Dickens). For a long time, the dodo remained a mystery. The few available primary sources came from travelers' accounts and from pictures of questionable accuracy. Until the mid-19th century, the only physical remnants were piecemeal: a head and a foot in Oxford, another foot in London, a skull in Copenhagen, and a leg and a piece of beak in Prague. Then, in 1865, the naturalist and schoolteacher George Clark discovered more than 300 dodo bones in a marsh in southeastern Mauritius, and in 1904, the only nearly complete dodo skeleton was found in a cave near the mountain of Le Pouce.

The study of the Le Pouce skeleton, along with other skeletal remains, has prompted a rethink of the dodo's traditional image as a sort of clumsy giant pigeon, a reputation deriving from one of the most famous representations of the bird in a painting by Roelandt Savery from around 1626. It is possible that the Dutch painter portrayed an overfed specimen from among those taken to Europe as curiosities. In the wild, the dodo was likely a leaner creature, consistent with sketches made *in situ* in Mauritius.

The dodo is now among the species that the "de-extinction" genetic engineering company Colossal Biosciences is working to simulate.

—Jordi Canal-Soler



The Twisted History of Barbed Wire

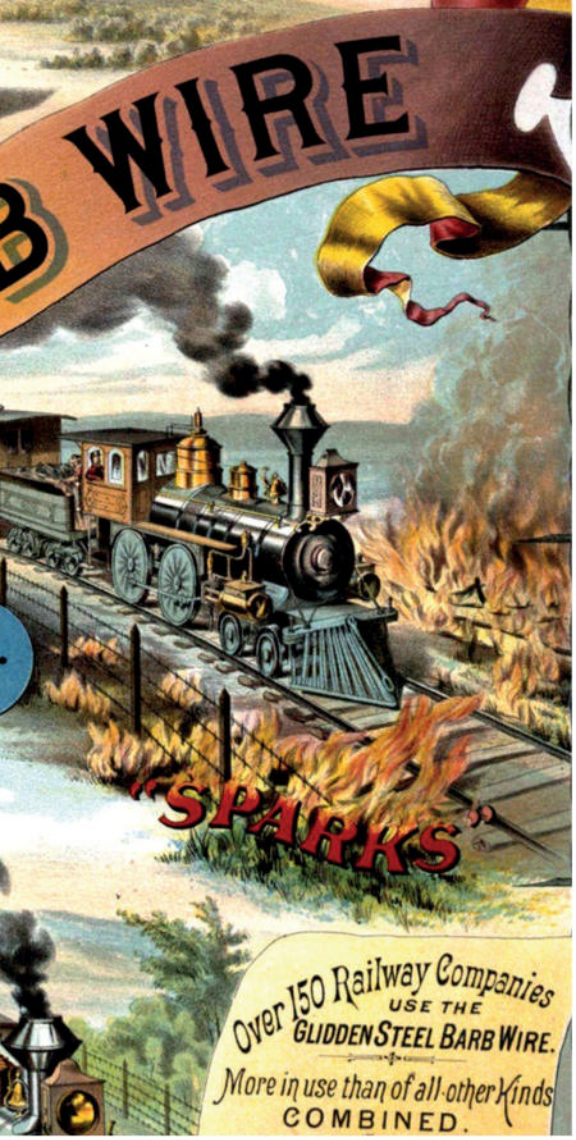
Since the innovative first patent was registered by Joseph Glidden in 1874, barbed wire has proved useful in many circumstances—to keep in, and to keep out.

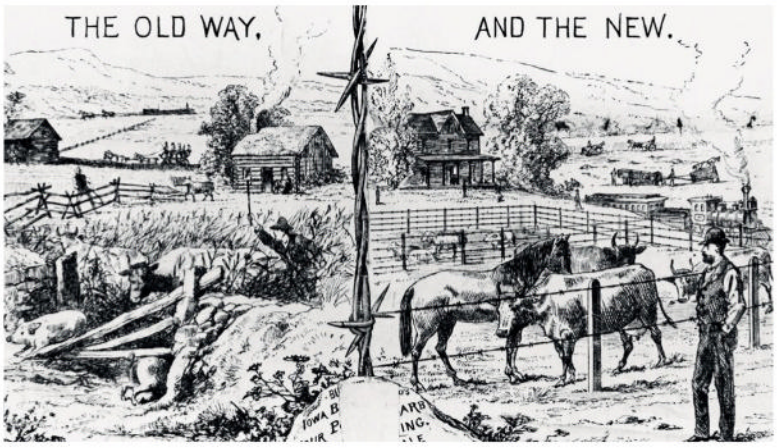
Indigenous people populated the Great Plains and prairies of the American West for thousands of years prior to the arrival of Europeans. For many Nations (such as the Sioux, Cheyenne, and Blackfoot), bison were central to their way of life, providing food, tools, shelter, and clothing. They migrated alongside the great herds, which once numbered in the millions. In the mid-1800s, as the devastation of the bison was underway, cattle and their attendant handlers (the American cowboys) arrived. The seas of grasslands that once fed the vast herds of bison also served as prime fodder for beef cattle.

Cattle and cowboys roamed where they pleased on the prairies of public domain. Cattle trails went from western Texas, north through the Indian Territory of Oklahoma into Kansas, up through Nebraska, and into the Dakota Territory. Unhindered by fencing, annual cattle drives moved herds to processing facilities conveniently located near settled railheads in Abilene, Omaha, and other towns.

Newcomers Arrive

The status quo shifted when President Abraham Lincoln signed the Homestead Act in 1862, which specified that any U.S. citizen could lay claim to up to 160 acres of land in the West. The only requirements were to build a home





ALL CLEAR ON THE RAILWAY

FROM ITS INTRODUCTION, barbed wire seemed a natural companion to the railroads beginning to crisscross the West. Rail lines cut across land previously reserved for livestock, and approaching trains would often strike the animals. Isaac Ellwood, who had purchased half the patent for Glidden's barbed wire, created a unique wire for railroad use. The motive for this design was to discourage wire theft. If railroad agents discovered a private field surrounded by railroad wire, the owner was taken to court for larceny.

An advert (left) for barbed wire shows a safely fenced-off railroad. Above, an illustration comparing life before and after the invention. LEFT, ALAMY; ABOVE, BETTMANN/GETTY IMAGES

there and to work on the land for five years. However, the lack of effective fencing limited farming and ranching practices—either to keep out the free-range cattle or to keep a farmer's own livestock contained. Wooden fences were costly, with lumber on the prairies in such short supply that homesteaders often built sod houses. Rocks for stone walls (common in New England) were likewise scarce on these plains. Shrubs and hedges took too long to grow and used water that was also in limited supply: The mid-19th century was a time of drought. Desperate for a solution, the homesteaders soon found one in what came to be known as "barbed wire."

The Development of Barbed Wire

Although a "wire fence" was patented in 1867, the idea for barbed wire as a method of fencing livestock started in 1868, when a man named Michael Kelly invented the basic design, which consisted of twisting two plain wires together to create a cable for barbs. Then in 1874, Joseph Glidden, a farmer based in De Kalb County, Illinois, received U.S. Patent #157,124, improving upon Kelly's invention. He also developed a way to mass-produce barbed wire, bringing down its price. Glidden's machine pulled two strands of wire tight around the barb, then wound the wires together around the regularly spaced spikes. As one satisfied user wrote: "It takes no room, exhausts no soil, shades no vegetation, is proof against high winds, makes no snowdrifts, and is both durable and cheap."

While solving the homesteaders' fencing problem, the introduction of barbed wire was most beneficial for large landowners, to the detriment of others. With the value of cattle rising, a land grab began, and landowners began to fence off vast swaths of land, including terrain that did not necessarily belong to them (often cutting off public roads and the best water sources). Indigenous people and the

Glidden saw a sample of barbed fencing at a fair in 1873 and began to devise improvements on it.

Joseph Glidden, ca 1877, owner of the patent of "Winner" barbed wire
LIBRARY OF CONGRESS/GETTY IMAGES

WIRE CUTTING VIGILANTES
This photo by Solomon D. Butcher shows a staged re-creation of barbed wire cutting by settlers in Nebraska: On closer inspection, it becomes apparent that the shears are fake.
BRIDGEMAN IMAGES

bison herds they followed could no longer move freely. Landless ranchers could no longer drive their cattle to market, nor could poor, small-scale farmers make use of the resources on the open range.

Indigenous people called barbed wire "the devil's rope," while the cowboys called it "the devil's hatband." Many despised the invention, which injured free-range livestock and bison, and ultimately took the lives of countless animals. Barbed wire's low-profile design makes the wire difficult to see, and it was easily overlooked by animals unfamiliar with the fencing. The barbs caused lacerations and subsequent infections, and animals would become entangled when attempting to jump over or crawl under the fences.

One result of the standoff between landowners and landless cattlemen was the Texas fence cutting wars, in which masked gangs such as the Blue Devils and the Owls went on wire cutting sprees, threatening fence owners to prevent them from rebuilding. One note read: "You are ordered not to fence in the Jones tank, as it is a public tank and is the only water there for stock on this range…No good man will undertake to watch this fence, for the Owls will catch him." The conflict ended primarily through legislative actions: In 1884, fence cutting was made a felony and illegal fences on public land were ordered to be removed. The amount of cutting declined, but it was heavy-handed enforcement

FENCED IN

IN THE POW CAMPS of the 20th century, barbed wire was omnipresent. Its spikes adorn the U.S. Prisoner of War Medal, which may be awarded to anyone detained in combat since April 6, 1917 (the date of the United States' entry into World War I). Cmdr. Everett Alvarez, Jr., was the first person to receive one, in 1986.

The U.S. POW Medal, encircled by barbed wire. GETTY IMAGES

THE DEVIL'S ROPE IN TIMES OF WAR

BARBED OR RAZOR WIRE would become a symbol of the inhumanity of war. During the Spanish-American War, the Rough Riders defended their camps with it, and the Spanish *reconcentración* fenced in the rural Cuban population to dire effect. Likewise, British troops used barbed wire in the Anglo-Boer War of 1899–1902, both for their own protection and to create concentration camps. During WWI, one expert estimated that by 1918, at least a million miles of wire had been strung through Flanders, Belgium, alone. One eyewitness stated that the wire on the western front was so thick in places that "daylight could barely be seen through it."

Soldiers in Russia tighten barbed wire in this colorized image from ca 1914 (WWI).

BRANDSTAETTER/GETTY IMAGES

by the Texas Rangers—even including one attempt to place bombs by the fences—that gradually put a stop to the practice.

The Growth of an Industry

Between 1867 and 1874, as many as 500 different types of spiked fencing were patented, with their inventors competing to improve effectiveness and capture a share of the market. As listed in *Early United States Barbed Wire Patents* by Jesse James (not to be confused with the outlaw), these fences were lined with "prods," "spurs," "prickers," "teeth," or "prongs." It wasn't until 1871 that Lyman P. Judson named his projecting metal points "barbs." The term stuck, though Judson's design did not survive.

Pioneers continued to tinker with the form, creating more than 2,000 unique homespun designs with inimitable names to match: the Underwood Tack, the Untorn Ribbon, the Necktie, the Hold Fast, and the Sunderland Kink, among numerous others. The most popular design, Glidden's, was named the Winner, due to the inventor's success in patent battles against competitors.

Today's barbed wire is highly durable. Manufacturers mix steel with carbon fiber for more flexibility, without compromising strength. Top-grade barbed wire is now coated in a mixture that is 95 percent aluminum and 5 percent zinc to prevent corrosion, keeping it in good condition for decades.

A Rusty Legacy

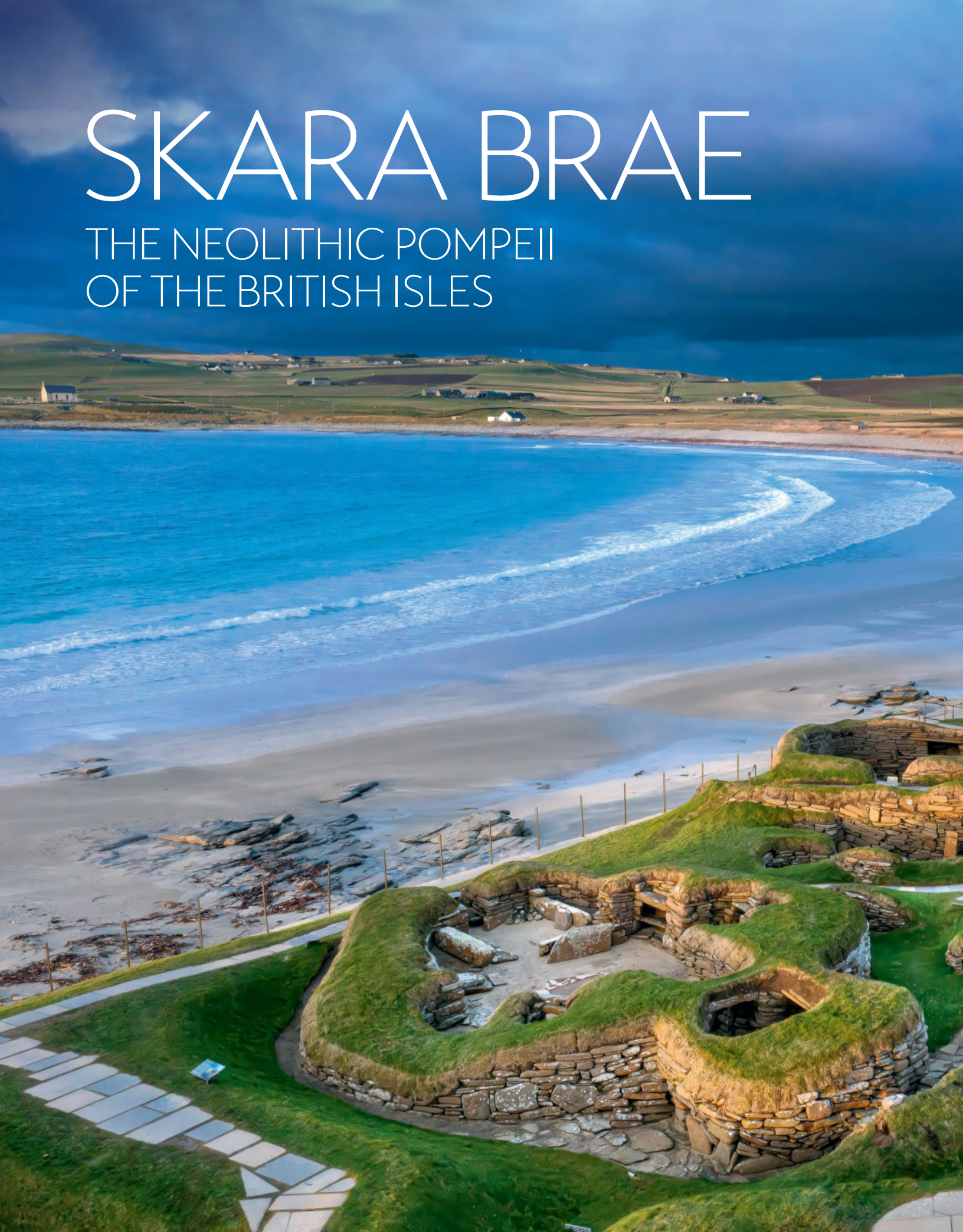
Barbed wire is ubiquitous across the world today, and its most common use (fencing for livestock) hasn't changed, despite its great versatility; it was even used as telephone wire in some rural areas before insulated copper wire arrived. Moreover, barbed wire's cultural resonance has hung on (quite literally) in the United States. There are barbed wire museums, competitions, and even an Antique Barbed Wire Society. There is also a growing movement of organized volunteer groups (such as the Barbed Wire Warriors of Colorado), who remove old barbed wire fences across the western United States to restore wildlife corridors for animals such as elk, deer, and moose—thus preventing injuries and opening up migration paths.

Along with the Colt pistol and the railroad, barbed wire is often listed as one of the three main technologies in "how the West was won." From the enclosure of the open range to the electrified fences of the Nazi camps, the 158-year legacy of barbed wire is well entangled in history.

—John C. Foltz

SKARA BRAE

THE NEOLITHIC POMPEII OF THE BRITISH ISLES



What did a Neolithic village look like? Magnificently preserved, the archaeological site of Skara Brae in Scotland's Orkney Islands transports us like no other to the daily life of farming communities 5,000 years ago.

DAVID V. CLARKE

WHERE THE ATLANTIC OCEAN MEETS THE NORTH SEA

The Skara Brae site, on the windswept coast of Orkney. The shoreline was farther away when the settlement was inhabited, but erosion has since brought it closer.

ALL IMAGES IN ARTICLE, UNLESS OTHERWISE INDICATED: MANUEL COHEN/AURIMAGES

ORKNEY
Pytheas of Massalia, the Greek explorer, passed by Scotland's northernmost tip, calling it "Orcas."

THOROUGH EXCAVATIONS
This photograph was taken at Skara Brae during the excavations led by V. Gordon Childe (below, at left). On the right are two of Childe's students, probably Margaret Simpson and Mary Kennedy.

A NEOLITHIC ISLAND SETTLEMENT

In the winter of 1850, an immense storm hit the Orkney Islands, which lie to the north of the Scottish mainland. At the Bay of Skaill, on the west coast of the biggest island, the storm stripped vegetation and sand from a large mound of a prehistoric midden more than 13 feet high. A midden is formed, for the most part, from the decomposition of organic matter, much like a gardener's compost heap, but here it was interleaved with layers of clay and sand, and contained ash, shells, broken bones, stone, and other waste from everyday activities. This was probably not the first time this mound had been exposed, but it was the first time that there were people eager to investigate it—primarily the family and acquaintances of the landowner, William Watt. Their work, partly recorded by George Petrie, a noted local antiquary, showed that the mound covered the remains of four houses lying either side of a roofed passage. Thus began the revelation of the best preserved prehistoric settlement in northern Europe.

An Archaeological Site

Once initial curiosity had been satisfied, Skara Brae saw only infrequent, small-scale, and poorly recorded investigations for more than 50 years. It was left to be filled by sand, blown in on the wind, and other debris. This remained the situation until 1925, when the descendants of William Watt put the ruins into the care of Her Majesty's Commissioners of Works. That winter saw a storm of the same intensity as the one in 1850. This immediate threat to the site's survival led the authorities to build a protective seawall in 1926. The cost of this construction prompted additional work to transform the site into a visitor attraction. Efforts to clear out the houses began in 1927 by a team of laborers under the supervision of an architect, but it quickly became apparent that a great deal of archaeological material had remained undiscovered by

3300 B.C.

The first settlement of Skara Brae was likely occupied around this date. Little is known about this settlement, or its inhabitants.

2900 B.C.

The second settlement is built. It comprises several one-room houses and is inhabited for around a century. Residents used Grooved Ware pottery.

2750 B.C.

Construction begins on the third and final settlement, which comprises six houses and a workshop and is abandoned around 2500 B.C.

1927-1930

V. Gordon Childe, one of the most distinguished prehistorians of the time, leads the excavations at Skara Brae, uncovering many additional artifacts.

NEEDLES AND MORTAR
Bone needles, possibly used to sew leather, skins, and animal hair to make clothing. Below, a mortar and grindstone for turning fish bone into fish meal for animal feed.

earlier investigators. So, between 1927 and 1930, the recovery of archaeological information was in the hands of Australian prehistorian V. Gordon Childe, author of the concept of "Neolithic revolution." This work yielded more houses and passages, together with many more artifacts. Finally, limited excavations took place at Skara Brae in 1972–1973, designed to uncover evidence about the site's environment and economy as well to collect as samples for radiocarbon dating.

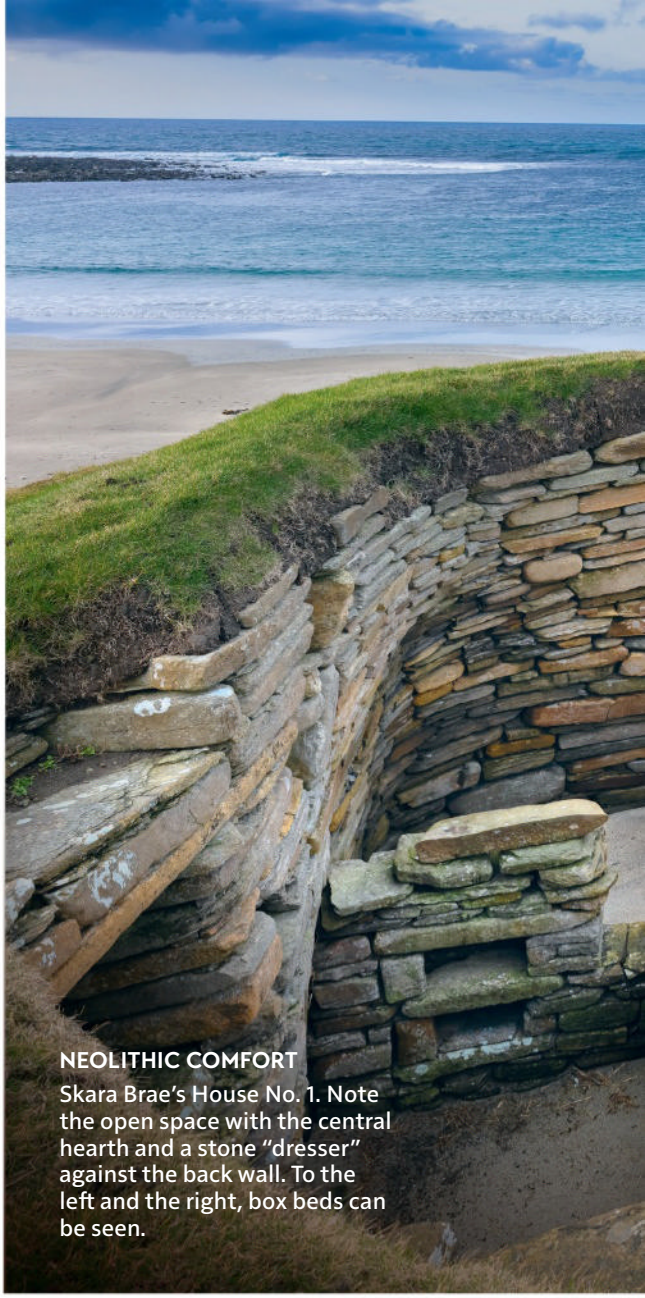
The Evolution of Skara Brae

These various excavations showed that there were three settlements, one on top of the other, at Skara Brae. The first and oldest, more than 5,000 years old, we know very little about, but it seems to have been abandoned at least 200 years before the next settlement was established, around 2,900 B.C.

This second settlement is comprised of one well-defined house and fragments of others, but in no case do the walls survive to more than a few courses high. These houses were freestanding, but clustered closely together. They all appear to have been built to a standard plan, consisting of a square room with rounded corners. The doorway was in the center of one wall, and opposite it was a stone "dresser." In the walls on either side of the door were bed recesses, and in the center of the room was a hearth. This settlement probably lasted less than a hundred years, but comparable settlements elsewhere in Orkney survived for several centuries. These settlements were built by groups who used a new form of pottery, known to us as Grooved Ware. This pottery style was invented in Orkney around 3200 B.C., and subsequently spread across Britain and Ireland. Unlike previous pottery styles in Britain, Grooved Ware used flat, as opposed to round, bases, a change that must have required new cooking techniques.

It is the third and final settlement, beginning about 2750 B.C. and ending just after 2500 B.C., that makes Skara Brae exceptional. Not only do the walls of some of the houses remain to roof level, but the major items of furniture survived, having been constructed from stone instead of wood. Nowhere else in northern Europe do we have this depth of information about what furniture might have looked like or how it was arranged in a house. We have this knowledge because of not only the near absence of usable timber but also the quality of the local standstone, which is easily split into rectangular slabs.

The core of this settlement was made up of six houses (Numbers 1, 2, 4, 5, 6, 7 as per the map on pages 30–31), linked by two roofed passages and a workshop (House No. 8). All but



NEOLITHIC COMFORT
Skara Brae's House No. 1. Note the open space with the central hearth and a stone "dresser" against the back wall. To the left and the right, box beds can be seen.

one of these houses, sharing their intermediate walls, flank the main passage, while House No. 7 is accessed down another passage running at right angles to the main passage. This dwelling may have been remodeled from a building belonging to the previous settlement, as its floor level is lower than that of the other houses. When the third settlement was reaching its end, these houses and passages were completely covered by a mound of midden, sand, and clay, which meant that their inhabitants were living a semi-subterranean life, well protected from the weather. The passages are not straight; their curves seem designed to throw the drafts of incoming cold air away from the doors of the individual houses.

Aerial view of Skara Brae. The structure on the far left corresponds to the so-called workshop, where stone tools were fashioned. Although it has a central hearth, the building stands out for its lack of box beds or a dresser. GETTY IMAGES

A Bird's-Eye View

The lack of readily available timber required ingenuity from the ancient settlers, who took advantage of natural walls and the heat of the earth to build their homes. This plan of the settlement was printed in the official guidebook to Skara Brae by the archaeologist V. Gordon Childe, published in 1950.

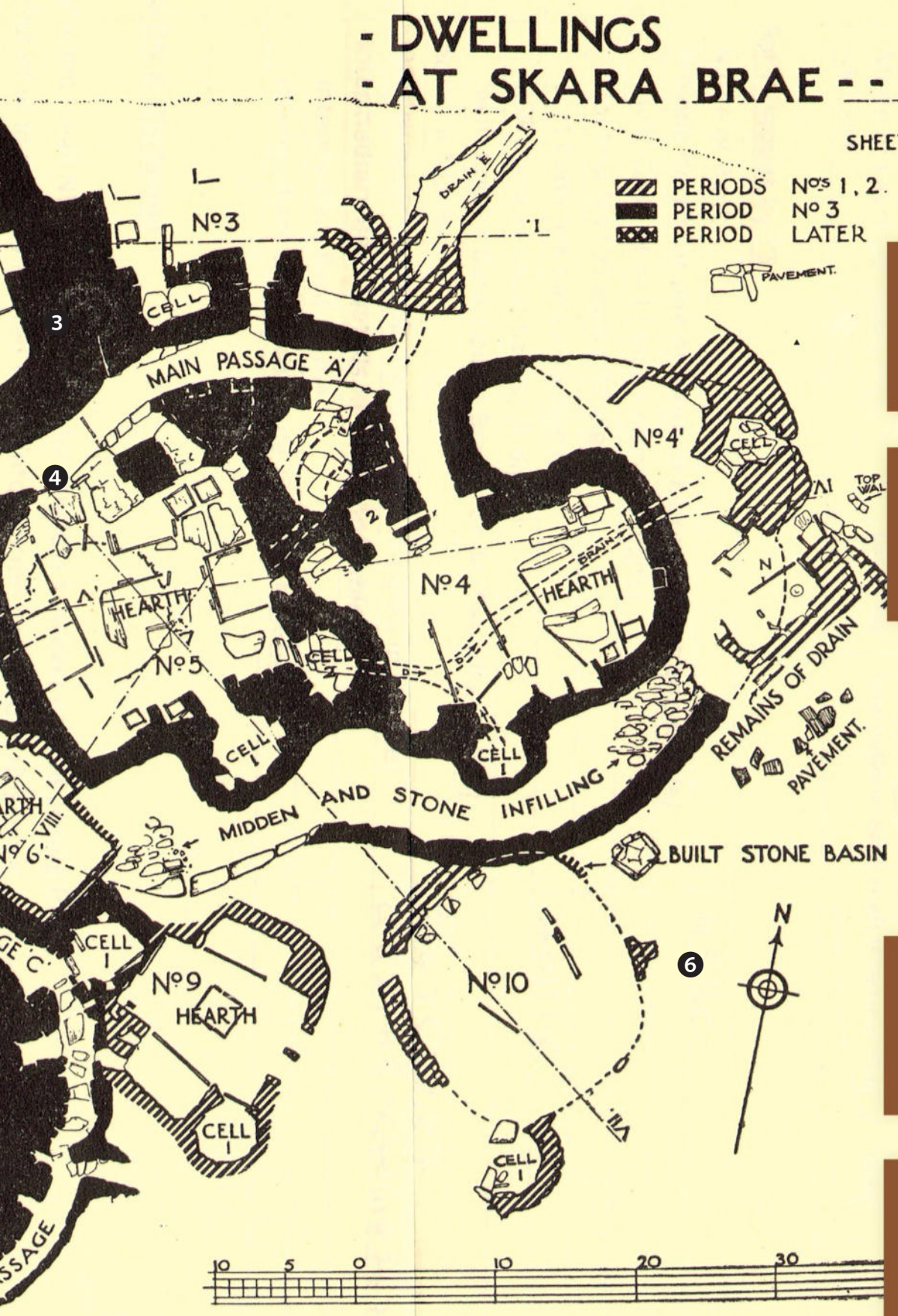
INTERNET ARCHIVE

1

Hearth
Each house comprised a single room with a floor area of about 430 square feet. In the center of the room, a fire would burn, used for heat and cooking.

2

Dresser
This was the main piece of furniture in the houses of Skara Brae. It may have been intended to display certain objects, or perhaps it had a ritual function.



3 Walls and ceilings
The walls are drystone. Given a lack of trees, the houses may have been roofed with whalebones, overlaid with turf, seaweed, or possibly stone.

4 Passages
These narrow covered structures would have allowed the inhabitants to move around while being sheltered from the wind and inclement weather.

5 Beds
Archaeologists found two rectangular stone boxes in each house, believed to be beds. These may have been filled with plants and skins to form a kind of mattress.

6 Entrance
The dwellings were entered through narrow passage-ways, some only two feet wide and four feet high. These were closed off with large stone slabs.

VESSEL AND AX
Above, a bowl made from whalebone. Below, a stone ax whose polished edges appear chipped due to use.

All these dwellings have similar layouts as those of the earlier settlement, but with some modifications and additions. The beds are no longer shallow recesses but are instead box beds, built of stone slabs, projecting from the wall. At the two corners of the front of the house stood stone pillars, probably supporting a skin roof and perhaps drapes for added warmth and privacy. Set into the floor in each house are a small number of stone boxes, their joints made watertight by clay seals. These boxes may have been used to soak limpets for use as fish bait. The thicker walls accommodate storage cells where valuable items could be safely stowed; one cell contained more than 2,400 beads. A storage recess is set into the wall above each bed. One type of cell, always situated in one of the two corners farthest from the door, has a high ceiling and large entrance. Running from it, under the floor, is a drain, part of a network of drains discharging beyond the limits of the settlement. It seems likely that these were indoor toilets, by far the earliest known in Scotland.

A GLIMPSE OF A NEOLITHIC DWELLING
Aerial view of House No. 1 at Skara Brae with its central hearth, box beds, and a large stone dresser set into the wall.

Under the Midden

The exceptional preservation of these dwellings has enabled observation of the dresser intact. This was perhaps the most important piece of furniture in the house. Two shelves, each formed of a single stone slab, are set horizontally into the wall. The upper slab is supported by a central stone block and by narrower slabs at either end. The lower shelf, itself above waist height, is kept in place by three pillars that rest on the floor. These shelves provide the only flat surfaces in the house other than the earthen floor, which is likely to have always presented irregularities. These dressers may have been simple storage units, or they could have provided a place for display, something like today's mantelpiece or sideboard, where a family's (or an individual's) prized belongings could have been shown. It is certainly possible that they had a more important symbolic meaning, acting as household shrines or altars.

The thickness of the house's walls causes their entrances to resemble small passages, unlike modern doorways. They are low and narrow, keeping heat loss to a minimum. Interestingly, this also means that, as you crawl through them, your immediate view is dominated by the central hearth with the dresser beyond. Between the hearth and the dresser is the only seat in the house, presumably reserved for the head of the household. The doors were not hinged; instead, halfway down

the entrance passage, there are two doorstops, one projecting from the ceiling and one from the floor. The door itself was probably a slab of stone, large enough to fill the entranceway. The door was pinned against the projecting stops by a bar, probably of whalebone, crossing behind it and fitting into slots in the walls of the entrance passage. This ability to lock the door was not intended as a security measure; these doors could be locked in place only from the inside. Perhaps largely a device to reduce heat loss, it also enabled the inhabitants to remove themselves temporarily from the wider, close-knit society.

With the door closed, these houses would have been very dark, lit only by the light of the central hearth. No lamps have been found.

The entrance to House No. 7 at Skara Brae, through which we can see the hearth in the middle of the dwelling, the seat behind it, and a stone dresser on the back wall.

PENDANTS AND OCHER
Above, bone beads and pendants. Below, a container for grinding ocher, which still harbors traces of the pigment that was used to decorate stones, skins, and fabrics.

In the absence of usable peat, so common today in Orkney, fuel was a mixture of animal dung, dried seaweed, heather, bracken, and marine mammal bone, rich in oil. This would probably not have provided sufficient light for detailed craftwork. Seeing the geometric designs carved on a few of the stones in the walls (even when colored in red, black, or yellow) would have been difficult. Apart from these stones, decoration seems to have been largely confined to the residents' pottery; any embellishment on bone or stone tools is rare. However, some of the inhabitants themselves must have been decorated by tattoos, as a small group of points, made largely from the wing bones of gannets, seem to have been used as tattooing implements.

Of course, not all decoration was intended to be permanent. Among the objects found were bone, stone, and shell vessels used to make red pigment from hematite, available on the adjacent island of Hoy, and possibly also on the Orkney mainland. Whether this pigment was used for body paint or for coloring the carvings, it would have soon faded.

Cooking and Crafting

The inhabitants of these houses were farmers, relying on animals rather than crops. Nevertheless, their diet was very varied. Their principal sources of meat (beef, mutton, lamb, goat, and occasionally pork) came from their animals. But these staples were supplemented by venison; ducks and seabirds, together with their eggs (including eider, golden plover, mallard, whooper swan, greylag goose, gannet, fulmar, puffin, guillemot, and great auk); fish (such as eels, trout, cod, salmon, haddock, halibut, and turbot); whale and seal meat; and shellfish (like crabs, mussels, oysters, and sea urchin). They probably collected wild plants, herbs, fruits, and nuts, but the only clear evidence we have is of hazelnuts and crab apples. Several fragments were found of rope made from the twisted roots of crowberry *(Empetrum nigrum)*. Ropes continued to be made in Orkney from this plant until well into the 19th century.

Many plants had medicinal use, best demonstrated by the significant number of puffballs *(Bovista nigrescens)* found at the site with their cottony inner tissue removed. This tissue helps to staunch the bleeding from wounds. The farmers

The inhabitants of Skara Brae depended more on animals than on crops. Their diet was based mainly on meat, fish, and wild plants.

knew their local environments intimately, both land and sea, and made full use of them.

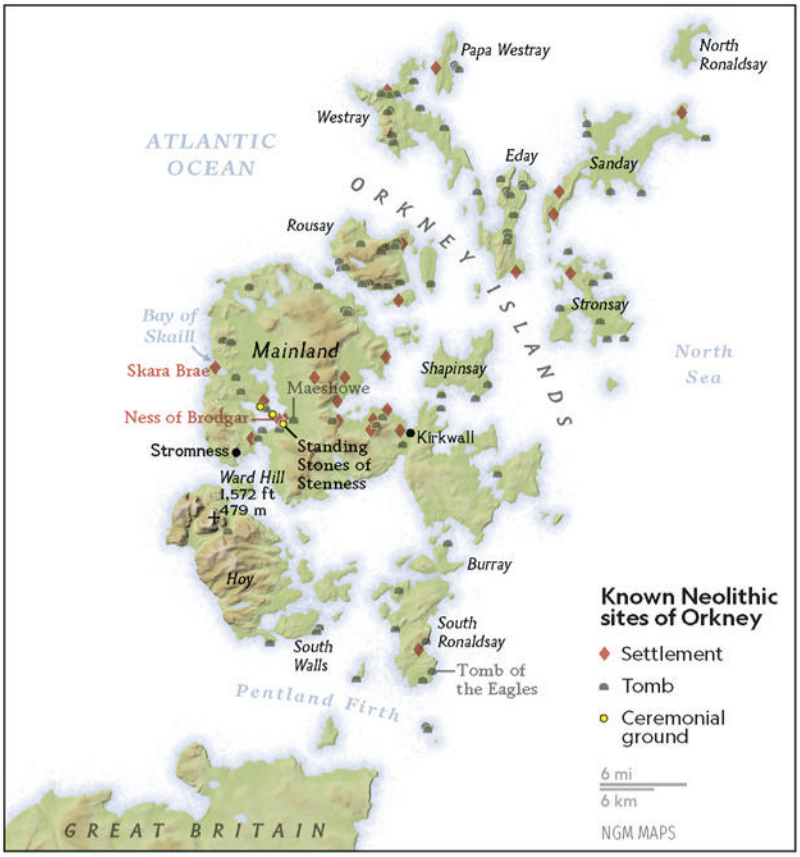
Standing beyond the houses is a workshop (No. 8). It has a central hearth but no furniture. When it was discovered, the floor was littered with fragments and tools made of chert (a quartz-based rock). Additional implements found at the site suggest that that items other than stone tools were made here.

Skara Brae is among the few settlements of early farming communities discovered in Britain with an extensive range of ornaments (pins, beads, and pendants), and these were certainly made at the site. These items, together with carved stone balls, mace-heads, and other unusual stone pieces, suggest that the farmers here were important leaders in the wider communities of Orkney. Skara Brae is also among the few settlements where many types of tools were made from bone. Most were likely used in the preparation of skins and furs for clothing. Strong, windproof attire would have been essential for surviving Orkney's harsh winters.

Orcadian winters make maintaining this site challenging. Climate change means that storms with the intensity of those in 1850 and 1925 will become more frequent. Coastal erosion at the Bay of Skaill offers an equally serious long-term threat, which is not restricted to Skara Brae. Many of the other sites in Orkney occupied at the same time as Skara Brae are coastal, and are suffering the same levels of erosion. ■

A LEADING EXPERT IN NEOLITHIC ORKNEY, **DAVID V. CLARKE** IS FORMER KEEPER OF ARCHAEOLOGY AT THE NATIONAL MUSEUMS SCOTLAND.

STANDING STONES
Two of the stones from the Ring of Brodgar, a Neolithic stone circle with burial mounds and henge, dating from around 2600–2400 B.C.



The Orkney Islands, a Major Neolithic Center

SKARA BRAE IS JUST ONE OF HUNDREDS of prehistoric sites that have been found in the Orkney Islands. The abundance of settlements here is not surprising: Despite the archipelago's remote location, it is home to fertile lands well suited to agriculture; the open landscape is ideal for livestock farming; and the climate is relatively mild considering Orkney's latitude, thanks to the Gulf Stream. Five thousand years ago, during the Neolithic period, these conditions were even more favorable. It is estimated that the islands were, at that time, home to a population of 10,000 people, half of Orkney's current population. An unusually large number of buildings have been preserved from the Neolithic settlements of Orkney because houses and temples were built using the easy-to-work outcrops of fine sandstone found there. The area richest in remains is the so-called Heart of Neolithic Orkney, which has been declared a UNESCO World Heritage site. In addition to Skara Brae, it includes the Maeshowe cairn, which contains a large burial chamber; the Ring of Brodgar, a structure similar to Stonehenge; and the Stones of Stenness, another stone circle. In 2002, an even more impressive site was discovered: the Ness of Brodgar, an enclosure with dozens of stone buildings, some of great size, found near the latter two sites on a narrow strip of land between the saltwater Loch of Stenness and the freshwater Loch of Harray. Structure 10, which measures 65 by 62 feet externally and has walls more than 13 feet thick, is the largest Neolithic structure of its kind in Europe. Findings suggest a specialized or industrial role, leading archaeologists to believe that the Ness of Brodgar could have been a ceremonial center that attracted communities from far and wide.

Maeshowe
This mound covers a huge burial chamber more than 5,000 years old. The entrance passage is perfectly aligned to receive the rays of the setting sun on the night of the winter solstice.

Ring of Brodgar
Potentially erected between 2,600 and 2,400 B.C., this circle, 340 feet across, originally consisted of 60 stones, of which 36 survive (only 27 standing), ranging from around seven to almost 16 feet in height.

Ancient Tools, Craft, and Magic

The inhabitants of Skara Brae used their ingenuity and the materials available to them to craft objects that made their daily lives easier, as well as others that may have been related to a magical and symbolic world.

Carved stone This unusual artifact may have been used in religious ceremonies or may have given its owner the right to speak in assemblies.

Whistle Flutelike object fashioned from a hollow bone with several holes drilled into it.

Symbolic object This carved stone item may have had a symbolic or ritual purpose.

Figurine Affectionately dubbed Buddo, this anthropomorphic figurine with eyes, nose, and navel, made from whalebone, dates to 2900–2400 B.C. and was discovered in a box bed in House No. 3.

Bone dice This whalebone carved with grooves and holes may have been used in divination rituals.

Beads
Beads made from animal bones and teeth were strung together to form necklaces and bracelets.

Decorative bone pin
This long bone with a hole in the center was probably used as a decorative pin. A string would have been threaded through the hole to hold it in place.

Pins and polishers
These bone objects were used to make holes and to sew leather garments. They may also have been used to polish leather and extract meat from crabs.

Pot
This re-creation of a complete Grooved Ware pot, similar to those used at Skara Brae in the Neolithic period, was based on a fragment of pottery discovered at the site.

Scraper
Although these items were traditionally thought to be hoe heads, new research suggests that they were scrapers for large hides.

MERITAMUN
A daughter and, later, a wife of Ramses II, Meritamun (whose name means "beloved of Amun") was a Singer of Amun who held titles including "Player of the sistrum of Mut and the menat of Hathor." Hurghada Museum. Opposite, a silver statuette of Amun merged with the god Re wearing his gold-plated feathered headdress. British Museum, London.

SANDRO VANNINI/BRIDGEMAN/ACI
RIGHT: SCALA, FLORENCE

THE PRIESTESSES OF AMUN

During the New Kingdom, ceremonies at the Temple of Amun involved female singers, dancers, and acrobats, organized under the supervision of the God's Wife of Amun.

ELISA CASTEL

THE OPET FESTIVAL
A drawing showing the moment when the three sacred barks (shrines in the form of miniature boats) of Amun, Mut, and Khonsu leave Karnak. The bark of Amun sports the figurehead of a ram.
BALAGE BALOGH/RMN-GRAND PALAIS

Herodotus claimed in his most famous work, *Histories,* that in ancient Egypt "no woman holds priestly office, either in the service of goddess or god; only men are priests in both cases." Yet a glance at the hieroglyphs, reliefs, and wall paintings of Egyptian temples and tombs quickly proves the Greek historian wrong. Women were, in fact, fully integrated into the service of the gods.

During the New Kingdom, at the height of Egypt's political and cultural power, the principal priesthood was devoted to the worship of the air god Amun at Thebes. Having previously merged with the sun god Re, he rose to prominence as "king of the gods," his singular and composite names used in varying contexts. His temple, the largest in Egypt, was called "The Most Select of Places" and is known today as the Temple of Amun at Karnak. The clergy was mostly male, and was led by four Servants of the God, including the high priest, who oversaw a range of subordinate officials of the deity within a clearly defined hierarchy. Alongside this male priesthood was a body of women, placed under the authority of the queen, the king's eldest daughter, or a woman to whom the function was delegated.

These priestesses, also known as Servants of the God, were typically drawn from daughters

ROYAL AND DIVINE WIVES

4th dynasty
ca 2543–2436 B.C.
Women enter divine service, mainly as priestesses in the pharaonic cult.

11th dynasty
ca 2080–1940 B.C.
The title of God's Wife makes a first appearance. At this stage, the title is purely honorary.

Sistrum depicting the face of Hathor, goddess of love, joy, and music. Egyptian National Museum, Cairo
BRIDGEMAN/ACI

GUARDIANS OF THE TEMPLE
Symbols of the god Amun, these sphinxes with the body of a lion and the head of a ram stand in the first courtyard of the Temple of Amun at Karnak, in ancient Thebes.

ALAMY/ACI

12th dynasty
ca 1939–1760 B.C.
The role of women in the religious ceremonies of the Temple of Amun declines.

18th dynasty
1539–1292 B.C.
At the start of the dynasty, Queen Ahmose-Nefertari is appointed God's Wife.

21st dynasty
1076–944 B.C.
God's Wife assumes control over a group known as the Singers in the Temple of Amun.

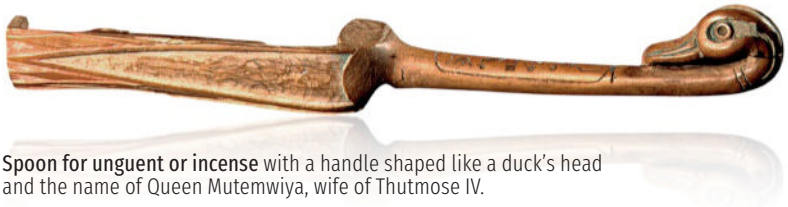
26th dynasty
664–525 B.C.
The office of Divine Adoratrice of Amun reaches the height of its political power.

THE WIVES OF THE GOD AMUN

A RELIEF IN QUEEN HATSHEPSUT'S Red Chapel at Karnak shows the God's Wife, probably Neferure (daughter of Hatshepsut, who had risen to pharaoh and thus passed this title to her), immersing herself in the sacred water of the lake at the Temple of Amun. Before performing her cultic duties, she purified her mouth with water and natron, a mineral salt mixture. The water from the Nile was believed to remove physical and spiritual impurities and allowed her to officiate without ritual contamination. The purification was also symbolic: It evoked the birth of the cosmic order that occurred at the moment of creation, known as *maat*. In the daily rites at Karnak, maat would be symbolically offered to Amun by the king or high priest.

PURIFICATION RITE This relief from the Red Chapel at Karnak shows the God's Wife at the temple lake, represented by the rectangle at her feet. The wave of water surrounding her figure indicates the act of ritual ablution.
UIG/ALBUM

and wives of the nobles and of temple personnel. Sometimes these women embodied goddesses such as Isis, Meret, or Mut, to whom ceremonies were dedicated. Their primary role, however, was to delight Amun through music and dance, as these were believed to be sacred performances capable of warding off evil. Presiding over these priestesses was the God's Wife. This office, which first emerged in the Middle Kingdom, was initially held held by the pharaoh's mother and later by the Great Royal Wife (the king's principal consort) and princesses. The God's Wife wielded considerable religious and magical authority, as she personified the lioness goddess Tefnut—the first female manifestation to emerge in ancient Egyptian creation myths, and was the deity charged with maintaining cosmic order, or *maat*. The title of God's Wife was also closely linked to the doctrine of divine kingship, since Amun was believed to manifest himself in the body of the pharaoh in order to impregnate his wife and engender the next ruler. The position of the God's Wife was of paramount importance during the New Kingdom. Endowed with substantial economic resources, she allowed the pharaoh to strengthen his ties with the powerful Amunite priests.



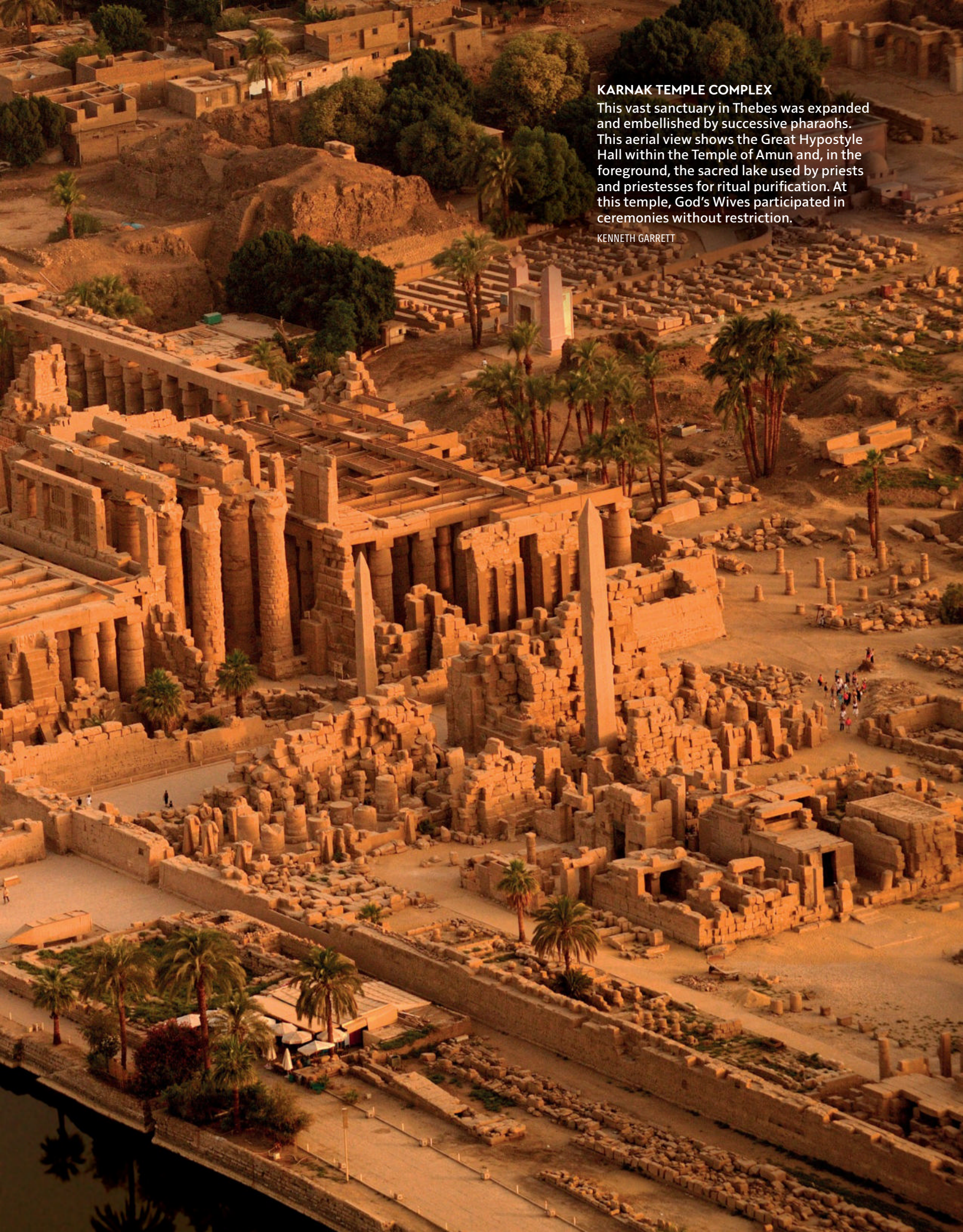
Spoon for unguent or incense with a handle shaped like a duck's head and the name of Queen Mutemwiya, wife of Thutmose IV.
AKG/ALBUM

Privileges of the Position

The first God's Wife to exercise these powers was Queen Ahmose-Nefertari, the wife of Ahmose I, founder of the New Kingdom. She was granted the honor of being appointed Second Priestess of the God by the oracle of Amun, a title never before held by a woman and one that formally integrated her into the leadership of the temple. This appointment was inscribed in the donation stela in which Ahmose I proclaims: "[I have given] the office of the second

AHMOSE-NEFERTARI, THE GOD'S WIFE
Great Royal Wife of Pharaoh Ahmose I, Ahmose-Nefertari was the first queen to hold the title of the God's Wife. She is shown here deified in a wooden statuette. Louvre Museum, Paris

CHRISTIAN DÉCAMPS/RMN-GRAND PALAIS



KARNAK TEMPLE COMPLEX
This vast sanctuary in Thebes was expanded and embellished by successive pharaohs. This aerial view shows the Great Hypostyle Hall within the Temple of Amun and, in the foreground, the sacred lake used by priests and priestesses for ritual purification. At this temple, God's Wives participated in ceremonies without restriction.

KENNETH GARRETT

A SACRED SITE
The Great Hypostyle Hall of the Temple of Amun features columns of up to 65 feet high and 16 feet in diameter. It was begun under Seti I and completed by his son Ramses II.
ALAMY/ACI

priest of Amun to the God's Wife, Great Royal Wife, united to the beauty of the white crown, Ahmose-Nefertari, may she live!" However, after this, the queen is no longer attested with the title of Second Priestess of the God and instead appears exclusively as the God's Wife. This shift highlights the growing importance of the office as head of the body of priestesses, and suggests that the authority of the God's Wife increasingly lay in directing female cultic personnel rather than in her formal rank within the priesthood.

Ahmose-Nefertari's residence when she was the the God's Wife was located at Medinet Habu, the temple of Ramses III. The position held substantial wealth, sustained by donations from the priesthood of Amun and from the pharaoh and members of the royal family. These offerings were formally accepted by Ahmose-Nefertari at the time of her investiture, when she declared: "Indeed, I am satisfied with the price." In addition to these endowments, the God's Wife likely received further gifts on the occasion of major religious festivals. Her assets included not only her residence at Medinet Habu but also estates and agricultural lands throughout Egypt, herds of livestock, and an extensive personal treasury comprising precious metals, garments, cosmetics, wigs, food, and other valuable goods. Under her command was an administrator, the Overseer of the Granary of the God's Wife, as well as other officials, who were male.

In reliefs, the God's Wife is typically depicted wearing a short wig and a formfitting dress held up by two shoulder straps. Her name is sometimes omitted in these images—reliefs were iconographic rather than hieroglyphic and were therefore visual narratives—but emphasis was placed on her leadership, symbolic role, and individual identity within the power structure of the dynasty.

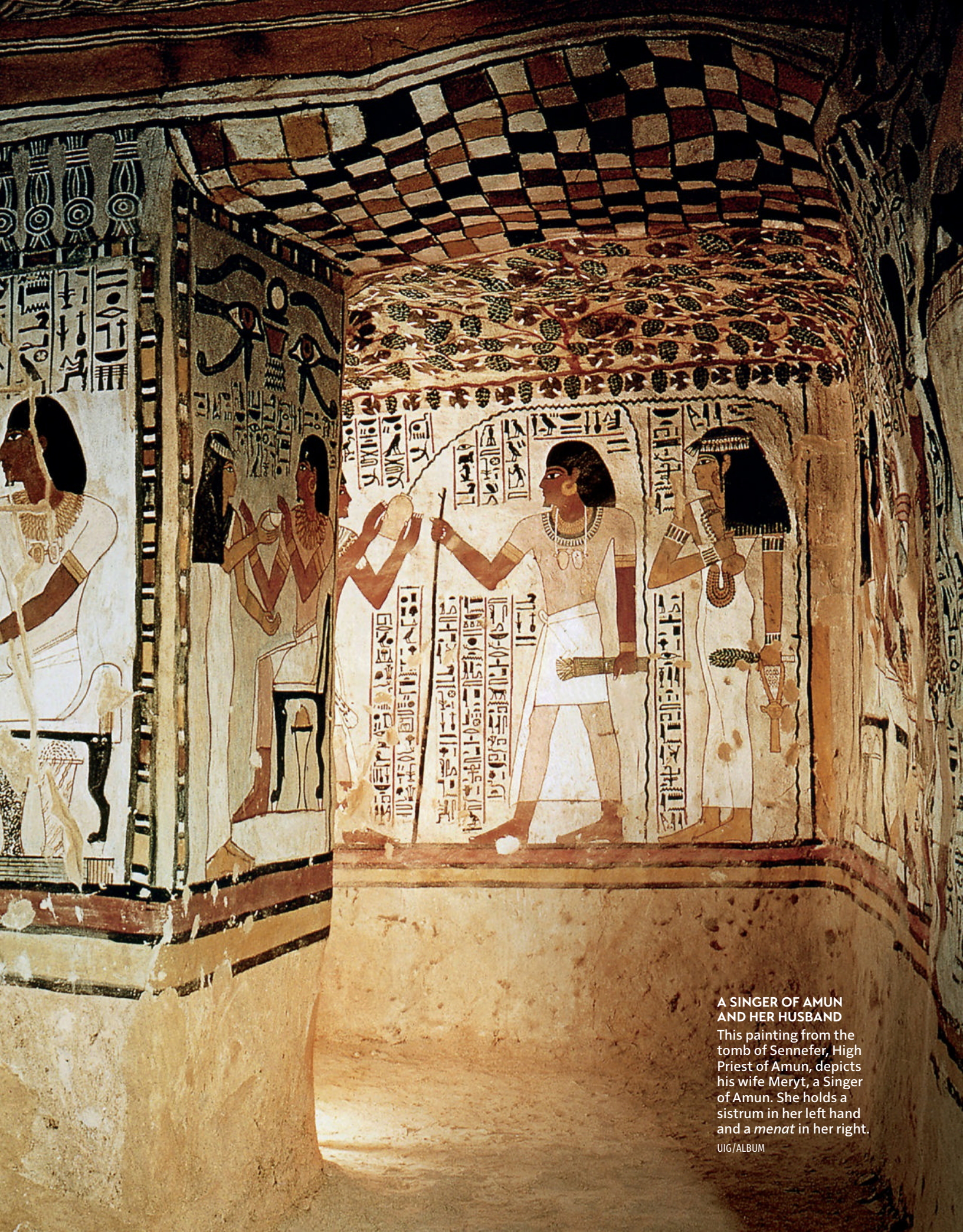
The deities Mut (at left) and Amun. Greywacke statue, probably from the Temple of Amun, Karnak. 13th century B.C.
DÉCAMPS/RMN-GRAND PALAIS

Singers and Musicians

The God's Wife delegated many of her duties to another woman of high rank, the Chief of the Musical Troupe of Amun. This office was the female equivalent of the first God's Servant, that is, the high priest or male official who oversaw the administration of the Temple of Amun. During the reign of Pharaoh Amenhotep III, the post was held by his mother-in-law Thuya; but upon her death it passed to Apeny, a woman outside the royal family and the wife of Ptahmose, High Servant of the God.

The Overseer of the Singers of Amun exercised authority over the largest group of Amun's priestesses, composed primarily of singers drawn from Theban high society, many of whom were daughters of priests. The positions were often held across generations, with daughters and granddaughters inheriting the same temple roles within a family. Upon their deaths, these women were buried according to the funerary customs of the elite: in richly decorated tombs, in polychrome coffins, alongside abundant grave goods and papyri intended to guide them through the afterlife.

These priestesses were organized into four groups, each serving for one month in a rotating cycle. Their ritual duties involved singing, music, and dance, and most bore the title of Singer of Amun. They performed in mixed ensembles and were accompanied by the *sistrum* and *menat,*



A SINGER OF AMUN AND HER HUSBAND
This painting from the tomb of Sennefer, High Priest of Amun, depicts his wife Meryt, a Singer of Amun. She holds a sistrum in her left hand and a *menat* in her right.
UIG/ALBUM

SONGS FOR AMUN

This relief from Queen Hatshepsut's Red Chapel at Karnak depicts singers and dancers—both men and women—taking part in the major festivals held in Thebes in honor of the god Amun.

ALBUM

① **DANCERS**
Two female dancers perform a gesture known as *nwn,* in which their hair falls forward covering their faces. The movement is associated with magical regeneration.

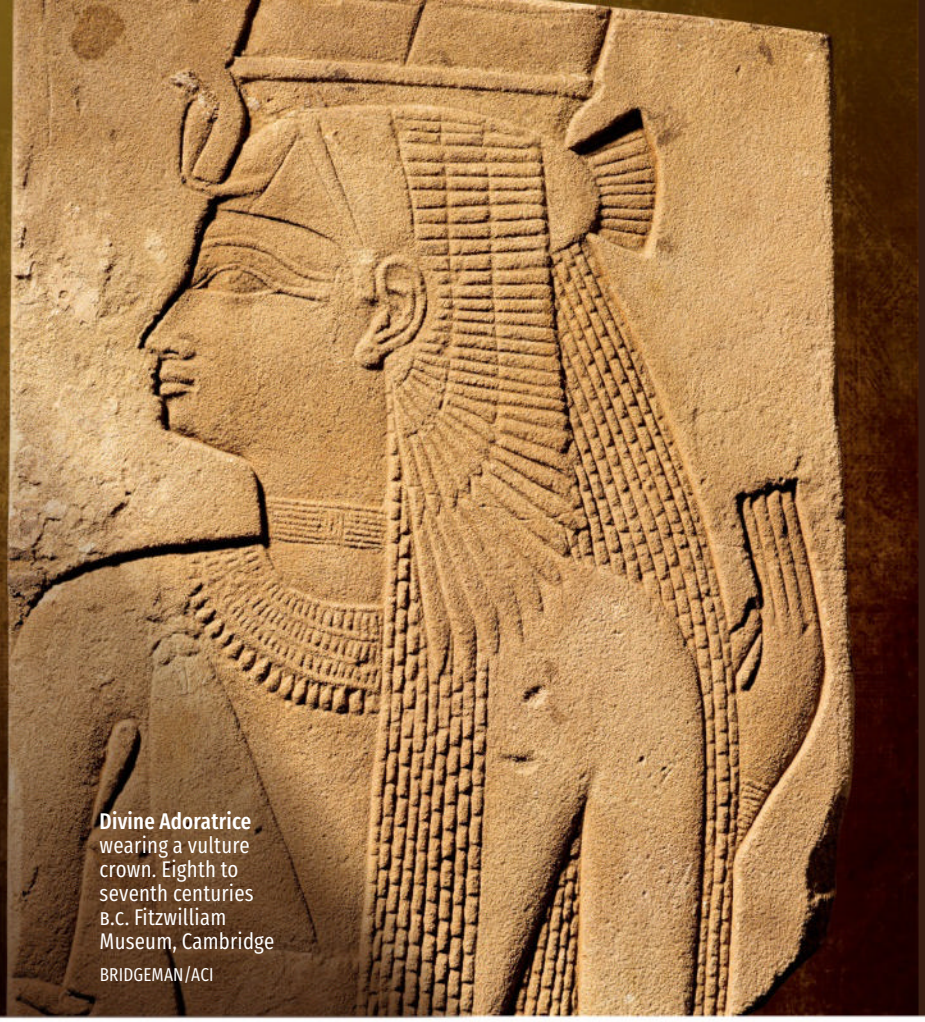
② **ACROBATS**
Six acrobatic dancers, wearing only loincloths, support themselves on their hands and feet while arching their torsos upward. The pose evokes the celestial vault.

③ HARPIST
A blind harpist and a singer perform a hymn to Amun. The harp rests on a stand, topped with the image of a small head of the pharaoh wearing the ceremonial crown.

④ DANCERS
Two male dancers, bare-chested and dressed in loincloths, perform a dance known as *iba*. The dance is recognizable by the position of their feet and arms.

⑤ CHOIR
A male choir sings in honor of the god, with arms raised and palms turned toward the sky. The singers are bare-chested and wear a longer skirt than the dancers.

⑥ MUSICIANS
Three female musicians hold a sistrum (a curved frame crossed by metal rods) in their right hand, and a *menat* (a necklace used as a rattle) in their left.



Divine Adoratrice wearing a vulture crown. Eighth to seventh centuries B.C. Fitzwilliam Museum, Cambridge
BRIDGEMAN/ACI

THE DIVINE ADORATRICE

AT THE END of the New Kingdom, a new female office appeared within the cult of Amun: the Divine Adoratrice. Over time, this role became more political than religious, and was often bestowed upon God's Wives or princesses. At the end of the 20th dynasty, there was a civil war. Power was later split between the pharaohs (who ruled from Tanis, the new capital in the northern Nile Delta) and the priesthood of Amun, who dominated Thebes in the south. The new office was created to mediate between the two. Elevated from an honorary title, the woman taking on this office had to be the unmarried daughter of the king, and would govern in Thebes on behalf of the pharaoh. In a period of political fragmentation across Egypt, the Divine Adoratrices functioned as both regents and high priestesses.

THE NAME OF A QUEEN
This cartouche bears the name of Nitocris, Divine Adoratrice of Amun. She took the office during the reign of Psamtik I, her father, in the seventh century B.C. Louvre Museum, Paris
DÉCAMPS/RMN-GRAND PALAIS

rattle-like instruments shaken rhythmically in time with their voices. Evidence suggests that this musical activity required formal training. A relief from the Middle Kingdom tomb of Khesuwer, a Supervisor of Servants and Instructor of Singers, depicts a figure holding two sistrums and then clapping his hands as he instructs a group of women. In addition to the priestesses, other musicians (male and female), took part in ritual performances. Reliefs and paintings show musicians clapping or playing a wide range of percussion, string, and wind instruments. These musicians had a crucial role in harnessing the protective power of music to ward off forces hostile to the god, forces that were thought to find such sounds offensive.

Dancers constituted another group of women attached to the cult of Amun. Among them the most prominent were the *meret* dancers (associated with the goddess Meret). Through their dance, they promoted fertility and took part in the rituals of Amun at Thebes. Some scholars suggest that both singers and dancers practiced a system of choreographic communication: Using hand gestures or facial expressions, they marked musical notes or dance positions for one another. The final group of priestesses associated with Amun were the acrobats. Most of these women came from elite social backgrounds, and were indispensable at festivals and ceremonial celebrations, where they performed sacred choreographies.

The singers, dancers, and acrobats were symbolically associated with sexuality and love, as certain dances and instruments—particularly the sistrum and the menat—were believed to stimulate the god. However, there is no evidence of a system of sacred prostitution as there was in the Near East and in Greece; priestesses were not required to engage in sexual activity as part of their cultic duties.

There is little doubt that the God's Wife and her priestesses played a central role in the religious and political life of the New Kingdom. Although men formally directed temple worship, the great ceremonies of the pharaonic religion would have been inconceivable without the Singers of Amun. ■

A MEMBER OF THE INTERNATIONAL ASSOCIATION OF EGYPTOLOGISTS, **ELISA CASTEL** IS A SPECIALIST IN ANCIENT EGYPTIAN RELIGION.

MUSIC FOR THE AFTERLIFE
This scene from the tomb of the scribe Amenemhat at Qurna shows a group of musicians including a harpist, a man playing a stringed instrument, and a woman playing a double flute.

S. VANNINI/DEA/ALBUM

SONGS FOR THE AFTERLIFE

Upon their deaths, the priestesses of Amun were regaled with splendid burials and funerary goods. It has been demonstrated that at least one singer, Sha-Amun-en-su, had her throat wrapped in resin bandages in order to preserve her voice beyond the grave.

PHOTOS: ALBUM

DENIUENKHONSU, A MUSICIAN OF AMUN (at right), makes offerings and a gesture of adoration to Re-Horakhty-Atum (a composite of all solar gods) in this wooden funerary stela. 950–900 B.C. British Museum, London

THE MALKATA MENAT, found near the palace of Amenhotep III. These necklaces were used as percussion instruments by the priestesses of Amun. Their soft, swishing noise was thought to appease the gods. ca 1390–1352 B.C.

MUMMY BOARD belonging to Henettawy (C), daughter of Isetemkheb, who bore the titles Mistress of the House, Singer of Amun, Chief of the Harem of Amun, Flutist of Mut, and God's Mother of Khonsu Henettawy. ca 990–970 B.C.

INNER COFFIN a chantress or Singer of Amun, also named Henettawy (F), ca 1000–945 B.C. MET, New York

The bravery and skill of these fighters drew thousands of spectators to

GLADIA

COURAGE IN THE FACE OF DEATH

THE FLAVIAN AMPHITHEATER
Built in the first century A.D., the Colosseum of Rome, with a capacity of around 50,000 spectators, was the largest amphitheater in the Roman Empire.

ANTONINO BARTUCCIO/FOTOTECA 9X12

amphitheaters across the Roman Empire.

TORS IN ANCIENT ROME

FERNANDO LILLO REDONET

When spectators took their seats in the stands of any Roman amphitheater, the moment they awaited most eagerly was the start of the gladiatorial combats. And they would have to wait: The day's events followed a set order, with animal hunts scheduled in the morning. These spectacles demonstrated Rome's dominance over the natural world and displayed the wealth and power of the games' sponsor, who could afford to import fierce and exotic animals from the farthest reaches of the empire. At midday, executions of criminals and fights between convicts took place. These events served as public reminders of Rome's justice system and its role in maintaining social order. The crowd's anticipation would only come to an end in the afternoon, when the gladiators finally entered the arena. From the terraces of the stands, the crowd watched as men displayed extraordinary bravery in the face of death.

Seasoned spectators would know their favorites by name, and would consult inscriptions that listed each fighter's past victories and the pairings scheduled for the day. Gladiators were highly specialized professionals; each underwent rigorous training to become a worthy opponent. Contrary to the modern popular imagination, it has been argued by historians that it was not so much the sight of blood that stirred the crowd, but rather a desire to witness a dramatic fight demonstrating the outstanding skill or courage of the combatants. In contrast to the wild chariot races at the Circus Maximus, where passions ran free, this was a comparatively stately event.

Gladiatorial Training

Gladiators were often prisoners of war, enslaved people, or criminals. Free men also entered the ranks, voluntarily, although they were in the minority and almost never from the upper echelons of Roman society. Whatever their background, all gladiators were rendered *infames*, a status denoting dishonor and the loss of legal and social rights. (Others subject to infamia included actors and prostitutes.)

DEATH IN THE ARENA

264 B.C.

The two sons of Decimus Junius Brutus Pera organize the first recorded gladiatorial combat in honor of their late father.

Fight between three pairs of gladiators, depicted in a mosaic from the third and fourth centuries A.D. Galleria Borghese, Rome

SCALA/FLORENCE

216 B.C.

The Roman Forum hosts funeral games in honor of Marcus Aemilius Lepidus. Twenty-two pairs of gladiators fight over three days.

65 B.C.

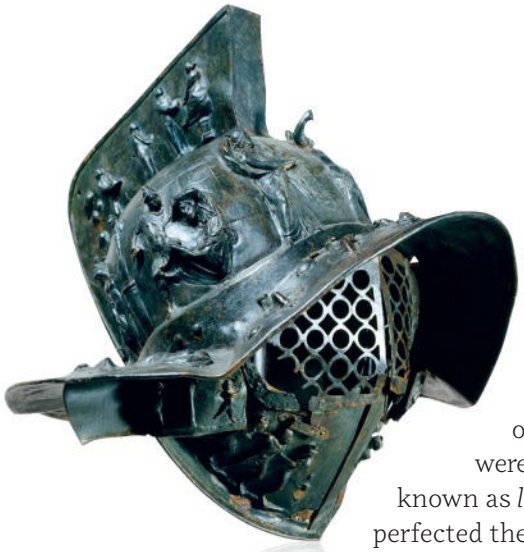
On the pretext of honoring his father, who died 20 years earlier, Julius Caesar stages large-scale games with 320 pairs of gladiators.

A.D. 80

Emperor Titus inaugurates the Flavian Amphitheater, later known as the Colosseum of Rome, with 100 days of games.

A.D. 523

The Colosseum hosts its final spectacle. By this time, the venue is used only for acrobatic displays and animal fights.



DEFENSIVE ARMOR
Helmet of a *murmillo* gladiator decorated with scenes from the conquest of Troy, found at the gladiator barracks of Pompeii. National Archaeological Museum, Naples
SCALA, FLORENCE

Deprived of recognized personhood, they were controlled by trainers, event sponsors, and politicians, and used for economic gain: They could be rented or sold, and all had a price. All were trained in gladiator schools, known as *ludi*, where they learned and perfected their fighting techniques. Such schools existed throughout the empire. The most famous, however, was in Rome itself, next to the Colosseum: The Ludus Magnus, whose remains are still visible today. The complex was connected to the Colosseum by a tunnel and contained its own smaller training amphitheater, with seating for around 3,000 spectators. The arena was a little over half the size of the Colosseum's: some 80 feet shorter and more than 40 feet narrower. Training there was the closest thing to fighting in the empire's greatest arena. The school could house hundreds of fighters in small cells set into three stories of porticoes that overlooked the arena. This was a highly diverse environment, filled with men from different backgrounds, speaking many different languages. Discipline was strict, though somewhat less so for gladiators of free birth. Some fighters had wives and children, yet it remains unclear whether their families lived within the *ludus* itself or had quarters nearby.

The trainers were known as *doctores* and each specialized in an array of gladiatorial fighter types, often being ex-fighters themselves. Although the exact details of training are unclear, one core exercise is well documented: sparring practice against the *palus,* a wooden pole that represented the opponent. Gladiators often trained with wicker shields and wooden weapons. These were not only blunt but also heavy, allowing fighters to develop the strength to wield a real weapon with greater agility. Gladiators could be well cared for and fed mainly on barley porridge, wheat, and legumes. They also received medical treatment from doctors, which included massages from specialists, as well as psychological preparation for combat. All of this was aimed at producing skilled professionals capable of satisfying the expectations of the most demanding of audiences.

Shield of a *hoplomachus* gladiator from the gladiator school in Pompeii. National Archaeological Museum, Naples
GETTY IMAGES

Fighting Specialties

Training and discipline at the ludi were overseen by the *lanistae*. In imperial schools, such as the Ludus Magnus, this role was filled by

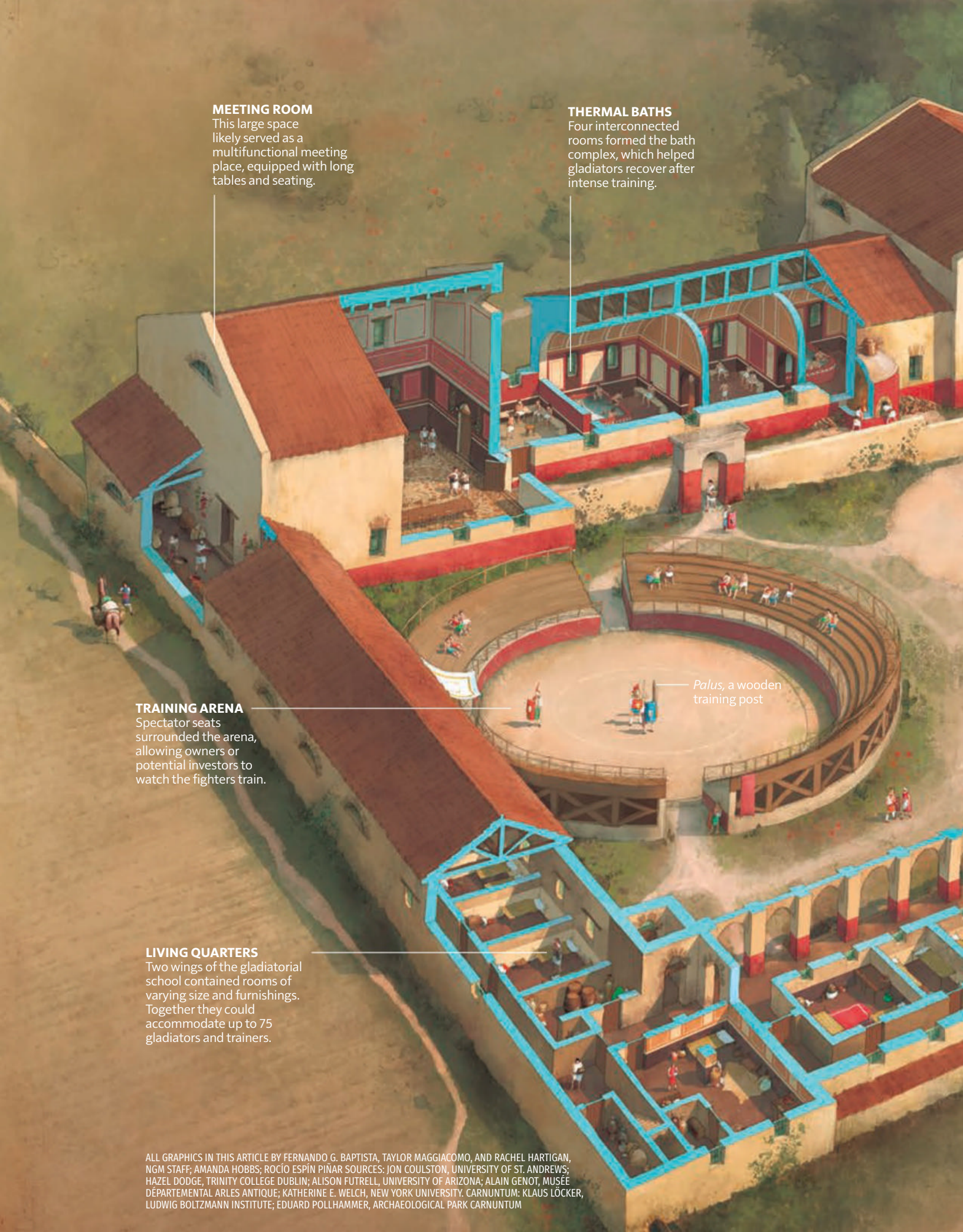
an administrator. Upon entering the school, gladiators were assigned a specific gladiatorial fighting type, or *armatura,* best suited to their physical abilities. They then perfected this role under expert supervision.

Many fighting profiles existed, but four were especially popular. The *thraex,* or Thracian gladiator, sported a closed helmet topped with the mythical figure of a griffin on the crest; a small rectangular shield; and two large greaves rising above the knees, which compensated for the small size of his shield. Equipped with a *sica* (a short, curved sword), he struck the opponent's unprotected areas. In combat, a thraex was usually matched up with a *murmillo.* The latter carried a larger rectangular shield, which allowed him to

AMPHITHEATERS ACROSS THE EMPIRE

ENTHUSIASM FOR GLADIATORS was not confined to Rome or even to Italy, as evidenced by the amphitheaters built throughout the Roman Empire. These buildings were designed to accommodate fans from neighboring towns who would come to attend the spectacle. In the western half of the empire, the landscape was dotted with newly constructed amphitheaters. In the east, however, new construction was less common, as existing theaters were more often adapted for gladiatorial use.

Sometimes rivalry between audiences led to serious clashes. This was the case in A.D. 59 in Pompeii, during Emperor Nero's reign, when Pompeians and spectators from Nuceria came to blows in the amphitheater, leaving many dead or injured. As a result, the Roman Senate banned the Pompeians from organizing gladiatorial games for a period of 10 years.



ALL GRAPHICS IN THIS ARTICLE BY FERNANDO G. BAPTISTA, TAYLOR MAGGIACOMO, AND RACHEL HARTIGAN, NGM STAFF; AMANDA HOBBS; ROCÍO ESPÍN PIÑAR SOURCES: JON COULSTON, UNIVERSITY OF ST. ANDREWS; HAZEL DODGE, TRINITY COLLEGE DUBLIN; ALISON FUTRELL, UNIVERSITY OF ARIZONA; ALAIN GENOT, MUSÉE DÉPARTEMENTAL ARLES ANTIQUE; KATHERINE E. WELCH, NEW YORK UNIVERSITY. CARNUNTUM: KLAUS LÖCKER, LUDWIG BOLTZMANN INSTITUTE; EDUARD POLLHAMMER, ARCHAEOLOGICAL PARK CARNUNTUM

GLADIATOR SCHOOL

In 2011, a *ludus*, or gladiator training school, was discovered in Carnuntum, a Roman city and military base in what is now eastern Austria. The site revealed details about how gladiators lived and trained. This reconstruction shows the layout of the school, which included two training areas and housing for dozens of fighters.

LEG PROTECTION
Greave with symbolic motifs. At top we see Bacchus, and at the base there is a stork fighting a snake. Found in Pompeii. National Archaeological Museum, Naples
LUCIANO PEDICINI/ ALBUM

get away with wearing a single greave and a closed helmet that completed his defensive equipment. A murmillo attacked with a short sword held in his right hand and often used his shield as a weapon.

Another popular pairing pitted the *retiarius* against the *secutor*. The retiarius was unique in that he fought without a helmet. Instead, he wore a *galerus,* a metal shoulder guard that offered some head protection. His offensive weapons were a trident and a weighted net, along with a dagger for close combat—often used after the loss of the net. When the category of retiarii first emerged, they typically fought murmillos. It wasn't until later that they were matched with secutores. The secutor (pursuer) relied on the relentless pursuit of his opponent in the arena. He wore a smooth, closed helmet with only two small eyeholes that severely limited his field of vision. Breathing under the heavy helmet was also difficult, and prolonged combat could quickly lead to exhaustion or suffocation. He also carried a large shield similar to that of the murmillos, in addition to a short sword.

Based on vivid descriptions of these gladiatorial fighting styles, historians have deduced that most blows targeted the areas where a fighter was most exposed. In any case, the equipment, both defensive and offensive, was designed so that gladiators couldn't easily dismember or decapitate their adversaries.

The Show Begins

At the opening of the games, the gladiators took part in a parade known as the *pompa.* It is unclear if they performed any kind of salute to the audience or the games' sponsors. What is certain, however, is that the phrase "Hail, Emperor, we who are about to die salute thee!" was never uttered by the gladiators. Instead, it was recorded by Roman historian Suetonius as spoken by criminals who had been condemned to death in a mock naval battle on Lake Fucine hosted by Emperor Claudius in A.D. 52. A version of the phrase became well known largely thanks to artist Jean-Léon Gérôme's 1859 celebrated painting *Ave Caesar! Morituri te Salutant,* which set the groundwork for its later use in modern novels and films.

The fighting began after the pompa, and was mostly a matter of single combat, but occasionally, more fighters stepped out into the arena. In Suetonius's profile of Emperor Caligula, he recounts a group fight between five retiarii and five secutores in which all the retiarii fighters were defeated. In an exciting twist, when the order was given to kill the defeated retiarii, one of them recovered his trident and killed all five victors. Another example of group spectacles involved so-called *pontarii* fighting: On a raised platform accessed by two ramps, a retiarius equipped with a trident, dagger, and projectiles had to prevent two secutores from conquering his position.

But what the crowd most eagerly awaited were the individual duels. Although the order of combat varied, some sources suggest that the *equites* (mounted fighters) were the first to perform. They likely began on horseback with lances and round shields, then continued their fight on foot, brandishing swords.

Each duel was supervised by referees who stood in the arena, close to the fighters. The *summa rudis* served as the chief official, often assisted by the *secunda rudis*. Both carried staffs as symbols of authority. They could halt the fight or grant a pause allowing the fighters a breather if they deemed it necessary. Their decisions adhered to rules that were familiar to both the gladiators and spectators but have since been lost to history.

Music was a vitally important component of the games. Mosaics and graffiti from the time confirm the use of various wind instruments, and a water organ (known as the *hydraulis*) may have also been used to mark timings and key moments in combat.

Combat Strategies

Gladiators adopted different fighting strategies depending on their specialty. In fights between a thraex and a murmillo, the latter

The phrase 'Hail, Emperor, we who are about to die salute thee!' was never uttered by the gladiators, but by the doomed participants of a mock naval battle.

SURROUNDED BY ADMIRERS
This detail from an 1891 oil painting by Stefan Bakałowicz, entitled *Gladiators Before Appearing on the Arena,* offers a romanticized view of the final moments before combat. State Russian Museum
FINE ART/ALBUM

FROM THE STREETS TO THE AMPHITHEATER

ROMAN FORUM (FIRST CENTURY B.C.)
During the Roman Republic, gladiatorial combats were staged in wooden structures temporarily erected in the Roman Forum, the civic heart of Rome.

10.000
spectators

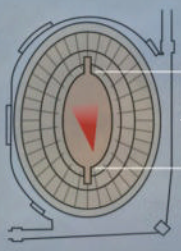
Referees
There were often two referees. They used staves to enforce the rules, the details of which are not known.

Provocator
These types of gladiators wore metal chest protectors and fought against each other.

Shields were used as both weapons and as defensive equipment.

These drawings depict two of the venues where gladiatorial games were held: the Roman Forum and the amphitheater of Pompeii. They also illustrate three types of gladiator: *provocator, murmillo,* and *thraex*.

POMPEII (FIRST CENTURY A.D.)
As gladiatorial combat grew in popularity, stone amphitheaters were built across the empire. Gladiators became increasingly specialized and their equipment became more elaborate.



22,000
spectators

Velarium
An extendable awning provided shade for the audience.

The word gladiator comes from *gladius,* a kind of sword.

Murmillo
A murmillo's weapons and armor could weigh between 35 and 40 pounds, and his large rectangular shield served both defensive and offensive purposes.

Thraex
Inspired by warriors from Thrace, this profile of gladiator carried a small shield and wielded a curved sword known as a *sica*.

SPARE WEAPON In addition to the trident, *retiarii* carried a dagger. This gladiator's dagger has an iron blade and a bronze and ivory hilt. National Archaeological Museum, Naples
SCALA, FLORENCE

relied on his large shield, using it as a wall to absorb the thraex's blows. These Thracian gladiators, by contrast, fought like birds with a beak. Armed with a short, curved sword, or sica, they circled their opponents in search of their weakest spots. The murmillos could remain protected behind their shield or strike back forcefully with it, pushing aside their opponent before attacking with a short, straight sword held in their right hand. Fights between retiarii and secutores offered an equally colorful and entertaining spectacle. The trident allowed the retiarius to keep his opponent at bay. The secutor, in turn, had to close in as quickly as possible before the restricted airflow inside his helmet caused exhaustion or suffocation. Retiarii used their nets to immobilize or unbalance their opponent, but the secutores' helmet was specifically designed so that the net would not get caught on it.

During the fights, spectators actively participated from the stands, shouting encouragement and praise. A *graffito* from Pompeii preserves some of these cries: "Fight, Ganges, Caesar is watching you!" while another goes so far as to claim: "You are the seventh wonder of the world!"

Death or Pardon

After a fierce battle, one of the pair may have been forced to surrender, either from exhaustion or because his injuries made it impossible to continue fighting. To signal defeat, gladiators might have thrown their shield or weapons to the ground and raised an index finger. At that point, the referees halted the fight and awaited the crowd's decision on the defeated gladiator's fate: death or mercy. In some cases, when both gladiators had fought with equal skill and neither had gained clear advantage, a shared pardon could be granted. The technical term for this outcome was *stantes missi* ("released standing"), meaning that they were free to leave the arena.

It's unclear exactly what gesture was used to announce the verdict, as no surviving ancient sources depict it visually. The expression *verso pollice,* cited by the satirical poet Juvenal to indicate a death sentence, simply means "turned thumb," without specifying the direction. The popular gestures of "thumbs-up" and "thumbs-down" spread, in part because of another famous Gérôme painting: *Pollice Verso,* painted in 1872. In the picture, the victorious gladiator looks toward

Some ludi included baths where gladiators cleaned themselves using *strigils* (scrapers) such as these found in Pompeii.
ALBUM

VICTORY AND DEFEAT This fragment of a first-century A.D. funerary relief shows a gladiator killing his defeated opponent with a sword. Lucus Feroniae Archaeological Museum, Capena

PLINIO LEPRI/AP PHOTO

the area where the vestal virgins sit, their thumbs turned downward in condemnation. This was not the only possible gesture: The audience may have signaled their decision by waving handkerchiefs or by shouting "*mitte*" to spare the defeated gladiator, or "*iugula*" to demand his death. In any case, the final decision rested with the patron of the games, who could accept or override the crowd's wishes. That said, he likely followed the spectators' verdict if he wished to secure their support and public approval.

In the Face of Death

It has been estimated that during the first century A.D., gladiators had about a 10 percent chance of meeting their end in any given fight.

THE DEATH OF A GLADIATOR

ONCE EXECUTED BY HIS OPPONENT, a gladiator's body was removed from the arena on a stretcher through the Porta Libitinensis, named after Libitina, the goddess of funerals. It was then taken to the *spoliarium*, the area of the amphitheater where weapons were stripped from the deceased. Burials would often be financed by the gladiator's *collegia* (burial society).

Dead gladiators were also the focus of powerful superstitions among the Romans. On the eve of their wedding, Roman brides had their hair parted with the tip of a spear—a weapon which, according to some accounts, had previously pierced the body of a deceased gladiator.

It was also believed that the still warm blood of a fallen gladiator, either collected in a cup or sipped directly from the wound, was an effective remedy against epilepsy.

THE GREAT SPECTACLE IN THE COLOSSEUM

By the second century A.D., gladiatorial combat had reached its height in terms of expense, scale, and spectacle. The Colosseum became the ultimate model for later Roman amphitheaters.

In some matches, a single *retiarius* faced two *secutores*. A platform with stones helped to even the odds.

A defeated gladiator signaled surrender by extending his index finger.

Secutor
Their weapons and armor weighed between 35 and 40 pounds and they wore a smooth, rounded helmet designed to avoid getting caught in the *retiarius's* net.

Retiarius
Carrying a maximum of 18 pounds worth of equipment, the retiarii relied on being agile and fast. They fought with a trident, a net, and a dagger.

Women, enslaved people, and foreigners
87 yards
Roman citizens
A galerus, or shoulder guard, offered partial protection for the head and neck.
Roman citizens
Nets over six feet high protected spectators from the action in the arena.
Other upper classes
Senators
Gladiatrix
Combat between female gladiators appears to have been rare. It remains unclear whether they fought with their chests exposed, as some statues appear to suggest.
Eques
These gladiators are thought to have begun their combat on horseback, armed with lances. They then dismounted to continue the fight on foot using swords.

When the death of the defeated gladiator was decreed, the execution followed immediately. Reliefs and Roman oil lamps suggest two common methods of putting an end to the defeated gladiator. One involved thrusting a sword through the gladiator's left collarbone until it reached the heart, resulting in a relatively quick death. The philosopher Seneca praised gladiators who faced death with courage, even guiding the victor's sword so that it would not falter: "So the gladiator, who throughout the fight has been no matter how fainthearted, offers his throat to his opponent and directs the wavering blade to the vital spot." Another method of finishing off an opponent required the defeated gladiator to get on all fours while the victor struck from behind, stabbing the gladiator beneath the left shoulder blade. In any case, the gladiator was expected to face death with courage. Cicero bears witness to this ideal when he asks: "Who that had been defeated ever drew in his neck to avoid the stroke of death? So great is the force of practice, deliberation, and custom!"

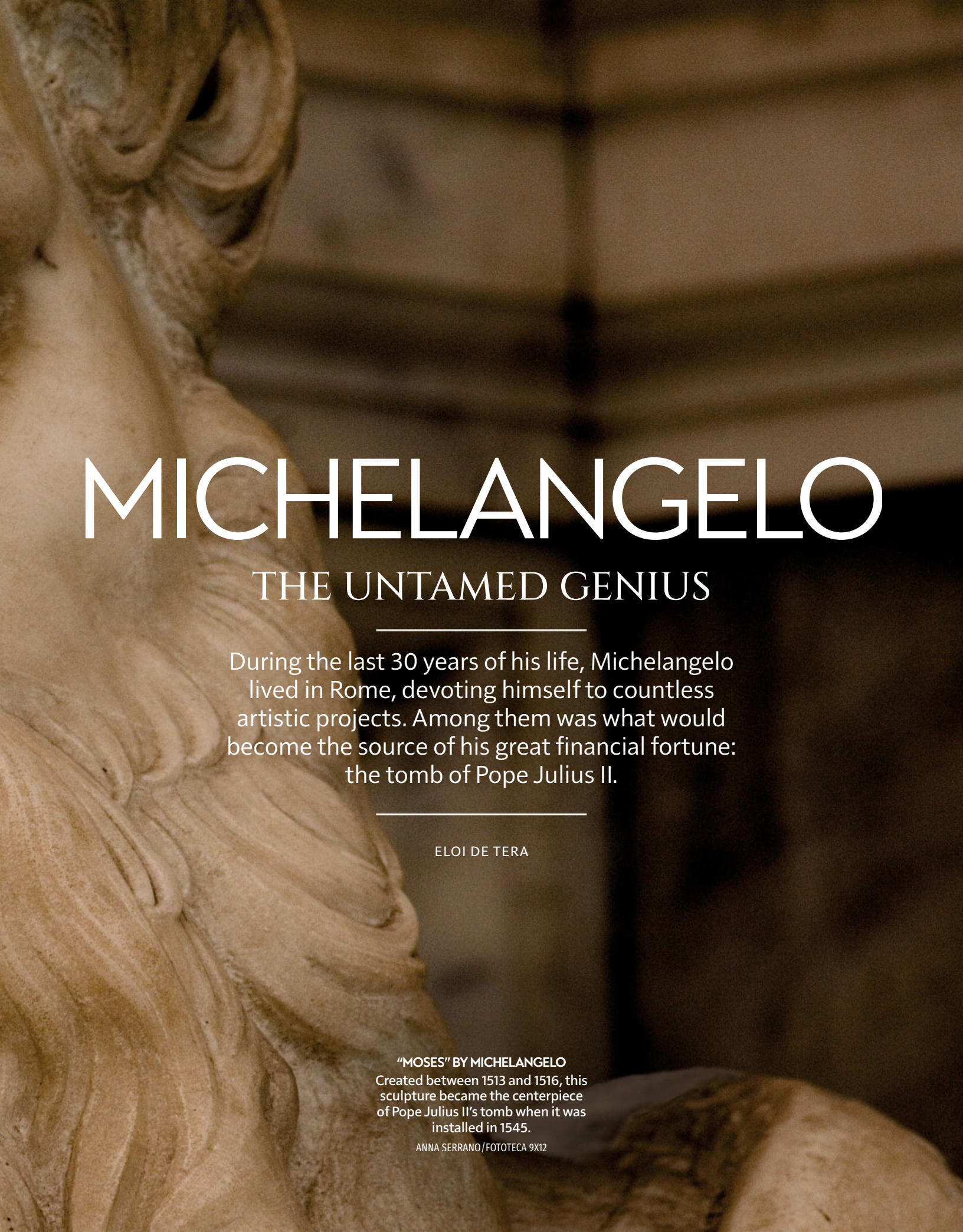
Afterward, the body was removed from the arena and funeral arrangements were made. Some gladiators would be forgotten, but the more fortunate received a dignified burial and even a memorial with an epitaph to preserve their memory and celebrate their victories. These were paid for by their family or by their fellow gladiators, often organized into *collegia* (unions that ensured proper burials). In some more privileged cases, the fighter himself, acting with foresight, may have set aside the necessary funds in advance.

Out of professional pride, many gladiators were reluctant to admit defeat. According to some surviving epitaphs written in the first person, they appear to have blamed their death on external factors. That of Diodorus reads: "Murderous Fate and the cunning treachery of the *summa rudis* [the senior referee] killed me, and leaving the light I have gone to Hades." An honorable end for an idol of the arena.

A LATIN TEACHER AND AUTHOR, **FERNANDO LILLO REDONET** HOLDS A DOCTORATE IN CLASSICAL PHILOLOGY.

UNDERGROUND WORLD Beneath the arena floor, the Colosseum contained an extensive underground network. It included passageways, cubicles used as cages for wild animals destined for the arena, and rooms for both the gladiators and the organizers of the games.

FRANK BACH/ADOBE STOCK



MICHELANGELO

THE UNTAMED GENIUS

During the last 30 years of his life, Michelangelo lived in Rome, devoting himself to countless artistic projects. Among them was what would become the source of his great financial fortune: the tomb of Pope Julius II.

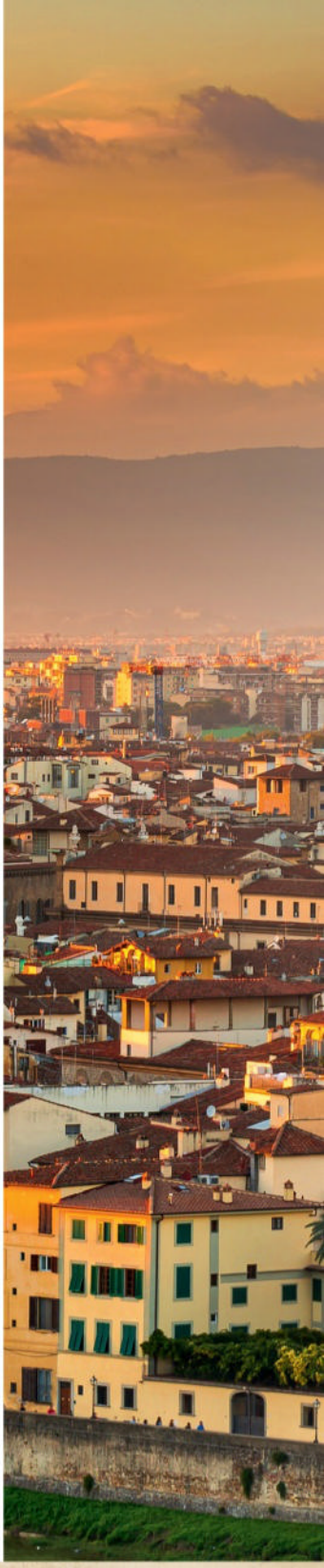
ELOI DE TERA

"MOSES" BY MICHELANGELO
Created between 1513 and 1516, this sculpture became the centerpiece of Pope Julius II's tomb when it was installed in 1545.

ANNA SERRANO/FOTOTECA 9X12

MEDICI CHAPEL
The tomb of Giuliano de' Medici (left), with figures "Night" and "Day," was designed and sculpted by Michelangelo.

GUIDO COZZI/FOTOTECA 9X12



THE ARTIST'S HOMELAND
This view of the city of Florence (right), shows the Arno River in the foreground and the Cathedral of Santa Maria del Fiore in the background.

SHUTTERSTOCK

The warm torpor of summer still lingered in the long afternoons of September 1534, when momentous news reached Florence. Giulio de' Medici, Pope Clement VII, had died on the 25th of that month, after allegedly eating poisonous mushrooms.

At the time, within the so-called New Sacristy of the Basilica of San Lorenzo in Florence (an architectural space conceived as a mausoleum for the Medici family), scaffolding was being erected under the watchful eye of Michelangelo Buonarroti. The artist, known to the world today as Michelangelo, was nearly 60 years old and had been commissioned to design the sacristy, which would mirror Filippo Brunelleschi's Old Sacristy. But events overtook him. A few days earlier, he had left Florence for Rome, where he arrived on September 23, just two days before the pope's death. Well acquainted with the inner workings of the Vatican, the news of Clement VII's condition may have convinced him that he should travel to Rome and be present when the white smoke rose above the roof of the Sistine Chapel, announcing the election of a new pontiff. Clement VII, Michelangelo's childhood friend and lifelong patron, had originally commissioned him to carry out two artistic projects that now hung in the balance: not only the

FROM THE ARNO TO THE TIBER

1501
After a stay in Rome, Michelangelo returns to Florence, where he will create his now iconic "David," a colossal statue symbolizing the Florentine Republic.

1505
Pope Julius II calls Michelangelo to Rome and commissions him to create an extravagant tomb in St. Peter's Basilica, where the pope's mortal remains will one day lie.

1534
After years in Florence, Michelangelo returns to Rome to build rapport with Pope Paul III, who makes him chief architect, sculptor and painter to the Vatican Palace the following year.

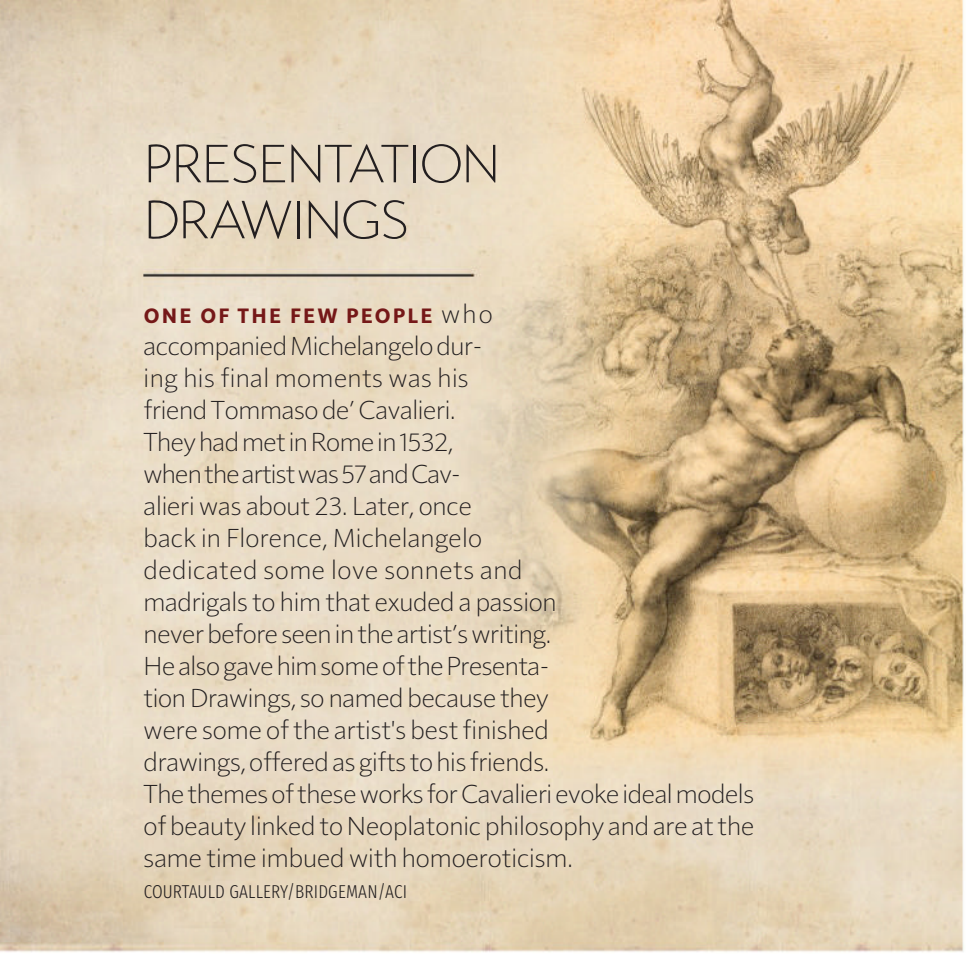
1564

Michelangelo dies at his home in Piazza Macel de' Corvi, Rome. Shortly afterward, his remains are transferred to Florence and buried in the Basilica of Santa Croce.

PRESENTATION DRAWINGS

ONE OF THE FEW PEOPLE who accompanied Michelangelo during his final moments was his friend Tommaso de' Cavalieri. They had met in Rome in 1532, when the artist was 57 and Cavalieri was about 23. Later, once back in Florence, Michelangelo dedicated some love sonnets and madrigals to him that exuded a passion never before seen in the artist's writing. He also gave him some of the Presentation Drawings, so named because they were some of the artist's best finished drawings, offered as gifts to his friends. The themes of these works for Cavalieri evoke ideal models of beauty linked to Neoplatonic philosophy and are at the same time imbued with homoeroticism.

COURTAULD GALLERY/BRIDGEMAN/ACI



UNIQUE DRAWINGS
"The Dream" (left) is one of the drawings Michelangelo made for Tommaso de' Cavalieri, around 1533. Courtauld Gallery, London.

GUIDO COZZI/FOTOTECA 9X12

THE SISTINE CHAPEL
The interior of the Sistine Chapel in the Vatican (right) shows frescoes by Michelangelo on the ceiling and altar wall.

SHUTTERSTOCK

unfinished mausoleum for the Medicis in Florence but also the splendid fresco of the "Last Judgment" that was to decorate the entirety of the Sistine Chapel's altar wall.

Although Michelangelo still had the strength to pitch and chisel marble without breaking a sweat, he was already an old man for his time, and 1534 had been a difficult year for him. His father Lodovico had died, and he had not seen the young nobleman Tommaso de' Cavalieri for many months. The two had met during the artist's brief stay in Rome in 1532, and it appears that Michelangelo had fallen in love. It was in this context that Michelangelo left Florence and headed for Rome, a decision that would prove momentous for his legacy.

The new pope, Paul III of the Farnese family, knew of Michelangelo's artistic talents, although he had never personally dealt with him before. In September 1535, he appointed Michelangelo chief architect, sculptor, and painter of the Vatican Palace, the pope's residence. The title came with a generous monthly salary from the papal treasury, the highest ever received by an artist of the time.

Pope Paul III, portrait by Titian, 1543. Capodimonte Museum, Naples

ALBUM

The task that most absorbed Michelangelo during the years that followed was the completion of the "Last Judgment" fresco in the Sistine Chapel. The task was exhausting, even after the upper part of the chapel wall was tilted forward to make the angle slightly easier to work with. Michelangelo's health was not as good as it had been 30 years earlier, when he had painted the vault of that same chapel. Then, in 1540, while painting the lower part of the wall, he fell from the scaffolding and had to spend more than a month in bed.

Commissions Throughout Rome

The Sistine Chapel was not the only project Michelangelo had taken on. In fact, over the years, he seemed to have left his mark in every corner of Rome. Modified under Michelangelo's direction, the new palace of the Farnese family (the family of the pope) was built near Campo de' Fiori square, not far across the river

ESAIAS
CONTVRBATIO-

LAURENTIAN LIBRARY
Michelangelo designed the vestibule of the Medici library (left), although the work was completed posthumously.
SCALA, FLORENCE

"CRUCIFIXION OF SAINT PETER"
On the right, the fresco by Michelangelo found in the Pauline Chapel of the Apostolic Palace in Vatican City. 1546–1550.
MASSIMO BORCHI/ OTOTECA 9X12

from Vatican City. He redesigned the Piazza del Campidoglio on Capitoline Hill, and the dome of St. Peter's also bore his touch. Meanwhile in Florence, without Michelangelo, who never returned there, the construction of the Laurentian Library continued according to his designs. Michelangelo sent back to Florence a clay model of what would be the library's monumental staircase. The artist controlled from a distance a team of laborers and artisans to complete the final touches on the sacristy of the Basilica of San Lorenzo. As if that were not enough, Pope Paul III wanted Michelangelo for a project in the Vatican: the Pauline Chapel. Among the countless major commissions Michelangelo had accepted on his return to Rome, he took on the most colossal project of all: the tomb of Pope Julius II. This Renaissance pope had been a renowned patron of the arts, commissioning major projects from the greatest artists of the time: Michelangelo to paint the Sistine Chapel ceiling; Raphael to decorate his private rooms and the three loggias of the Apostolic Palace; and the architect Donato Bramante to build a new St. Peter's in Rome to replace the existing basilica, which dated to the time of Constantine. The heart of this new building (the future St. Peter's Basilica) was to be marked by the tomb of Julius II himself, which he had commissioned Michelangelo to build decades earlier, in 1505.

Majolica bowl with the coat of arms of Julius II. Metropolitan Museum of Art, New York.
MET/ALBUM

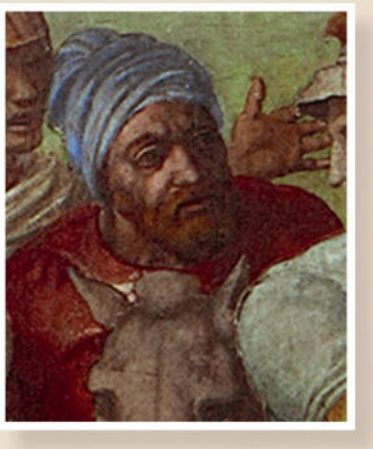
Personality Clash

The tomb had been one of Pope Julius II's big projects, and forever marked his relationship of artistic patronage with Michelangelo. Julius II had always believed in the brilliance of Michelangelo. That said, there were quite a few clashes between the two men.

The outrage Michelangelo experienced in 1506 when Pope Julius II refused to receive him has been well documented. On that occasion Michelangelo reacted by leaving for Florence,

COMMISSIONED BY PAUL III

MICHELANGELO'S OTHER CHAPEL



AFTER HIS ELECTION AS POPE, Paul III had a small, private papal chapel—known as the Pauline Chapel—built near the Sistine Chapel. Michelangelo painted two large frescoes here. Although it was easier to work in this space than in the Sistine Chapel, progress was slow: He took three years to paint the "Conversion of Saul" and four more years for the "Crucifixion of Saint Peter." By comparison, it had taken him around four years to paint the entire ceiling of the Sistine Chapel. But Michelangelo was tired and busy with the completion of Julius II's tomb. Although he was still finishing the apocalyptic "Last Judgment," with the Council of Trent already underway, the new pope did not allow him to paint any nudes. In the "Crucifixion of Saint Peter," Michelangelo portrayed himself, somewhat younger, wearing a blue turban, attire he was occasionally depicted wearing. In this self-portrait, rediscovered during the restoration of the fresco completed in 2009, we see Michelangelo's distinctive features and the same terrible gaze of his "Moses" also depicted in the face of St. Peter. Every pope knows when entering this private chapel that it's looking to him with attention as follower of the Petrine ministry.

SCALA, FLORENCE

ST. PETER'S BASILICA
Upon being appointed papal architect in 1546 for the project, Michelangelo drew up a Greek-cross plan for the new basilica in Rome, featuring an imposing dome that would not be completed until 1590.

DANITA DELIMONT/AWL IMAGES

Julius II examines the plans for St. Peter's with Michelangelo, Bramante, and Raphael. Oil painting by Horace Vernet, 1827.
BRIDGEMAN/ACI

determined never to return. The pope sent messengers and letters, writing to intermediaries such as the gonfalonier of Florence (an important city official), but nothing could change Michelangelo's mind. Julius II, called the warrior pope by historians, had already been conquering territory for the Papal States. Finally, having taken the city of Bologna, Julius II received Michelangelo there and commissioned him to create a large bronze sculpture depicting him as a conqueror. Michelangelo accepted the task, but the work was destroyed when the Bentivoglio, the lords of Bologna, recaptured the city.

Expensive and Unfinished

The tomb Michelangelo originally designed for Julius II was colossal, a four-sided monument that a person could walk under and into, covered with sculptures on every side. It was almost as if he wanted to compete with the legendary Mausoleum of Halicarnassus, one of the Seven Wonders of the Ancient World. Its towering architectural structure was to be divided into three levels, intended to house more than 40 marble sculptures, as well as some bronze reliefs. The monument was to be erected in the center of the new Basilica of St. Peter's, then still under construction. The precise plan was for Michelangelo's tomb for Julius II to be installed directly under the great dome (also by Michelangelo), where sculptor and architect Gian Lorenzo Bernini's baldachin now stands.

The sculptures proposed for the mausoleum included representations of figures such as Moses, St. Paul, and, of course, St. Peter. These figures were designed to evoke the two sides of Pope Julius II: a pope who led the papal armies into battle, and a humanist prelate and patron of the arts and letters.

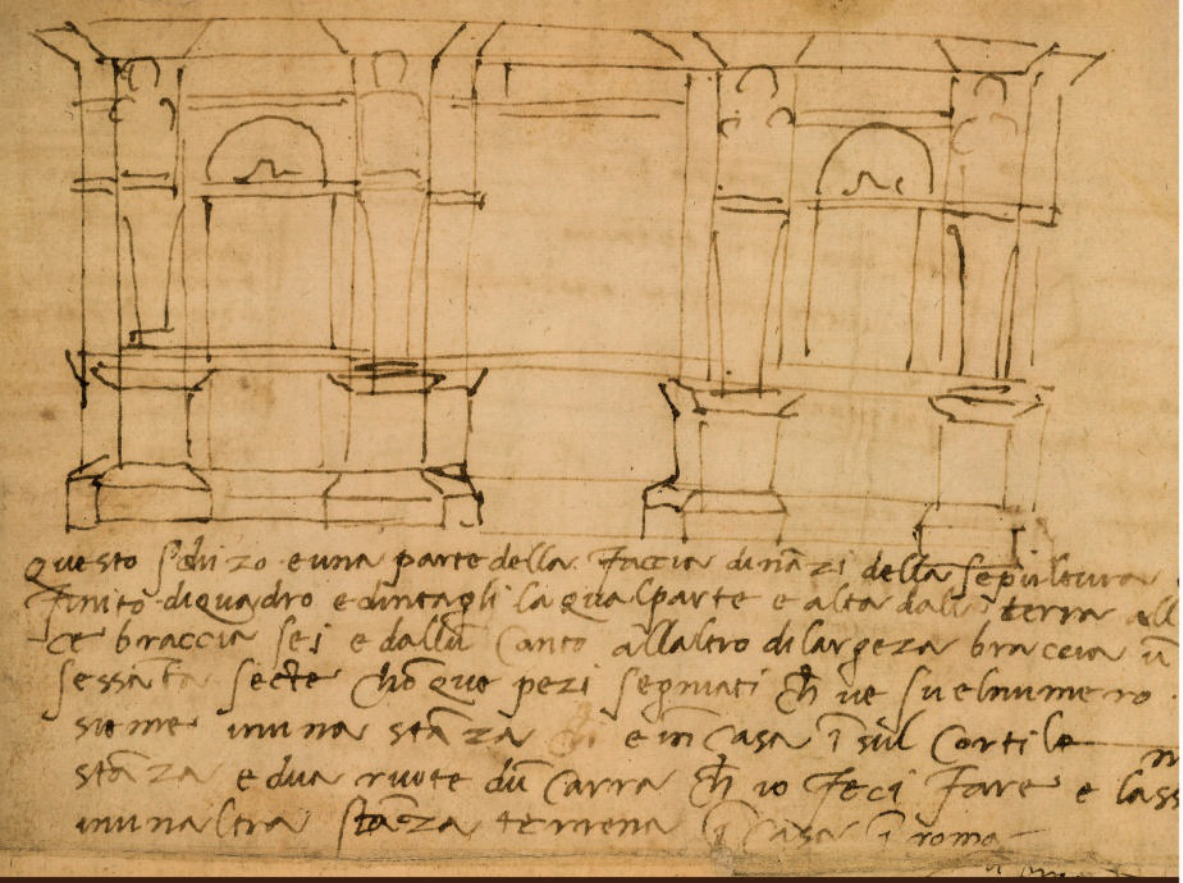
This project dogged Michelangelo throughout his life. After the death of Julius II in 1513, he signed successive contracts with the deceased pope's heirs in which the number of sculptures decreased, while the price increased. Michelangelo was given a house in Rome's Piazza Macel de' Corvi (now demolished) where he lived for much of his life and where he stored all the marble blocks he had purchased for the mausoleum project. But from 1516 onward, he did not touch a single one of those blocks.

In this context, Michelangelo was being paid for doing absolutely nothing. He received more

THE PIAZZA DEL CAMPIDOGLIO

Pope Paul III commissioned Michelangelo to redesign this square in Rome, and ordered the equestrian statue of Emperor Marcus Aurelius to be moved to its center. Michelangelo also designed the Palazzo Nuovo (in the background), which was not built until the 17th century.

LUIGI VACCARELLA/FOTOTECA 9X12



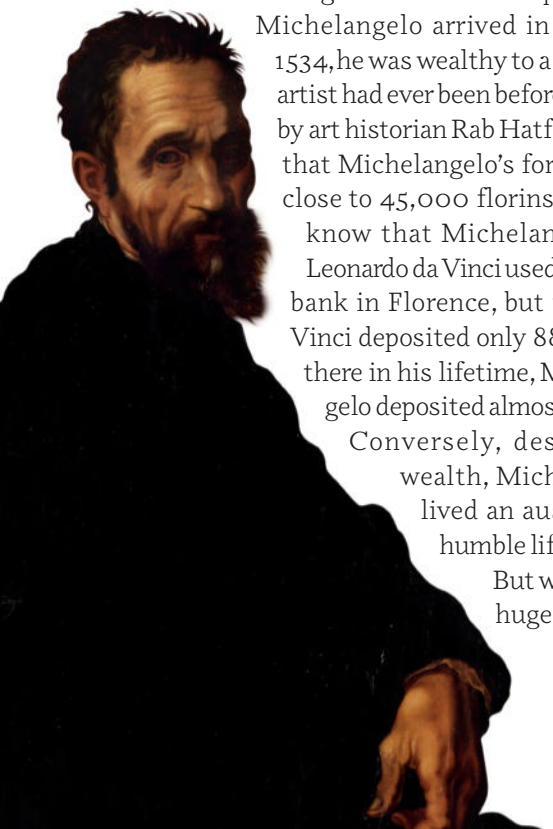
Sketch of the lower story of Julius II's tomb, 1518. British Museum, London BRITISH MUSEUM/SCALA, FLORENCE

than 4,063 florins in payments connected with the project—more than for any other commission in his life. By comparison, he was paid 2,300 florins for the facade of the Basilica of San Lorenzo in Florence (which was never completed), and only 3,000 florins for painting the ceiling of the Sistine Chapel. When Michelangelo arrived in Rome in 1534, he was wealthy to a degree no artist had ever been before. Studies by art historian Rab Hatfield show that Michelangelo's fortune was close to 45,000 florins. We also know that Michelangelo and Leonardo da Vinci used the same bank in Florence, but while Da Vinci deposited only 889 florins there in his lifetime, Michelangelo deposited almost 10,000. Conversely, despite his wealth, Michelangelo lived an austere and humble life.

But what was a huge financial deal for Michelangelo ended with the worst possible artistic result. Had he known, Julius II would have been outraged. The grandeur of the mausoleum that Michelangelo conceived during the pope's lifetime and which was to be the focal point of the new Basilica of St. Peter's was hugely diminished. The poorly finished facade, sparsely adorned with sculptures, expressed none of the extraordinary iconographic significance of the initial project. Thanks to Paul III, Michelangelo obtained permission to complete the project with the help of assistants, but out of everything he had already crafted between 1513 and 1516, and which was stored in his house in Macel de' Corvi, he included only his sculpture of Moses in the final ensemble.

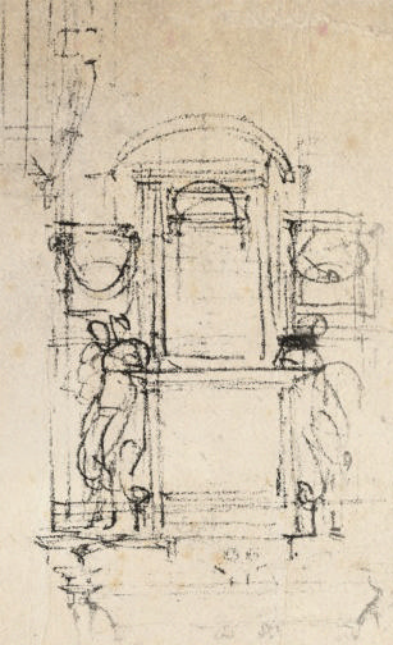
The Great Disappointment

The combination of funding being redirected for other papal projects and Michelangelo's commitment to his other commissions proved fatal to the ambition of the artist's initial designs. The tomb with more than 40 sculptures that was to preside over the new St. Peter's Basilica was relegated to the minor basilica of



Michelangelo ca 1540. Portrait by Jacopino del Conte. SCALA, FLORENCE

TOMB OF JULIUS II
The plan for a colossal tomb for Julius II was ultimately reduced to a monumental facade on one side of the Church of St. Peter in Chains in Rome. In 1545, the monument was completed with sculptures by Michelangelo's collaborators.
SCALA, FLORENCE



THE EVOLVING PLANS FOR THE TOMB

There were at least six contracts by Michelangelo for the mausoleum of Julius II; sketches of some of them are preserved, such as the reproduction above in the British Museum.

ALBUM

THE TOMB THAT COULD NEVER BE

IN THE EARLY DESIGNS for the tomb of Julius II, such as the sketch from 1513 (right), the focus is clearly directed at the figure of the pope. Two angels lift him from his sarcophagus, representing his resurrection at the moment of the Last Judgment Ⓐ. At his feet appear eminent figures, including Moses Ⓑ, while on the lower level are allegories of Charity Ⓒ and Faith Ⓓ. Michelangelo planned to cover all the columns with statues of naked men, in the manner of caryatids Ⓔ. The figures adopt poses inspired by the "Laocoön"—a sculptural ensemble found in Rome but thought to be Greek, in which the Trojan priest and his sons are attacked by giant serpents. According to Ascanio Condivi, Michelangelo's biographer, the sculptor's naked figures (below) were allegories of the arts that, after the death of the pope (their great custodian), had been imprisoned. This also explains their name "prisoners" (the name "slaves" came much later). Some were left unfinished, and none made it into the final project at the Church of St. Peter in Chains. This may be because their nudity and eroticism ran contrary to the conservatism that was, by then, taking hold in the Church at the time of the Council of Trent.

❶ "Rebellious Slave" made around 1513–1515. Louvre Museum, Paris
RAPHAËL CHIPAULT/RMN-GRAND PALAIS

❷ "Bearded Slave" unfinished, ca 1530–1534. Galleria dell'Accademia, Florence
AKG/ALBUM

❸ "Genius of Victory" ca 1532–1534. Palazzo Vecchio, Florence
WHITE IMAGES/SCALA, FLORENCE

❹ "Dying Slave" created ca 1513–1515. Louvre Museum, Paris
SCALA, FLORENCE

Second project, from 1513, for the mausoleum of Julius II. Copy by Giacomo Rocchetti. State Museums, Berlin

BPK/SCALA, FLORENCE

THE LAST SCULPTURE

IN HIS FINAL WORK, the "Rondanini Pietà," Michelangelo repeated the same exercise he had already carried out in his "Moses:" converting one sculpture into another. From an earlier, larger version of the statue, made in the 1550s, Michelangelo began to carve another smaller sculpture with a changed position, continually modifying its layout. But it would remain unfinished. On February 12, 1564, Michelangelo worked on the piece for the last time, the same day he fell mortally ill; he passed away six days later. It is still possible to see elements of both versions in the unfinished "Rondanini Pietà," now preserved in the Sforza Castle, in Milan. This work provides us with a snapshot of a process of transformation, of the artist's working method, which is still contemplated today.

PRELIMINARY SKETCHES
A study sketch of the "Rondanini Pietà" (left), ca 1552, is the only clue to how the finished sculpture should look. Ashmolean Museum, England
BRIDGEMAN/ACI

AN IMPOSING MOSES
Giorgio Vasari wrote of Michelangelo's "Moses" that in the delicate rendering of hair "it seems as if the chisel has become a brush."
SCALA, FLORENCE

St. Peter in Chains, where it was placed on one side of the transept. Strictly speaking, it is not a tomb but a cenotaph: a simple funerary monument, as Julius II is actually buried in St. Peter's Basilica under a plain marble tombstone. The "tomb" displays only one sculpture made in its entirety by Michelangelo—the imposing figure of Moses. The other sculptures on display were made almost entirely by Michelangelo's collaborators, while many of the sculptures that Michelangelo had already carved for the tomb (such as the nude "slaves," now in the Louvre) remained unused in his workshop.

Retouching the Masterpiece

"Moses" is the centerpiece of the ensemble, and the statue was already completed in 1516. In 1542, as the day approached to transfer the work from his home to the Basilica of St. Peter in Chains, Michelangelo had second thoughts about the work.

Until then, Moses had been looking straight ahead, and his beard had also been positioned at the front. But following the comments of a friend who saw the statue in his home, the artist took up his hammer and chisel and began to chip away at Moses' head to turn his face to the left. This fact is recorded in a letter from the period and has been verified by the historian and restorer Antonio Forcellino.

This was an incredible feat that only Michelangelo could achieve: twisting a finished marble head, also reworking the torso and a leg. Today this alteration is almost imperceptible, seen only in the turn of the beard as it reaches the hands and connects with the only fragment of beard that corresponds to 1516, as well as in the back of the neck, which is abruptly turned.

Michelangelo died in his studio in Macel de' Corvi on a cold February day in 1564. Just six days earlier, he had been working on the "Rondanini Pietà." Initially, he was given a burial in Rome, but when the news of his death reached Florence, Michelangelo's former compatriots set out to claim his body. One of the artist's nephews, Leonardo Buonarroti, secretly whisked away his remains, defying those in Rome who wanted to keep him buried there. In accordance with his wishes, Michelangelo returned once and for all to Florence, where, after a princely funeral in the Basilica of San Lorenzo, he was buried in the church of Santa Croce. ■

ELOI DE TERA IS A SPECIALIST IN RENAISSANCE AND ITALIAN MANNERISM AND THE ART OF THE QUATTROCENTO AND CINQUECENTO.

The Midas Mound: Full of Treasures, Empty of Gold

In 1957 archaeologist Rodney S. Young excavated a gargantuan burial mound with an inner chamber that offered a series of surprises.

The Indo-European Phrygians created a flourishing kingdom between the 12th and sixth centuries B.C. in central Anatolia (modern-day Türkiye). Its capital was the citadel of Gordion (then Gordium), located about 45 miles southwest of where Ankara now stands. Over 120 burial mounds dot the area surrounding the ruins of Gordion. The Phrygians entombed members of their elite inside these barrows, or tumuli—clearly visible today.

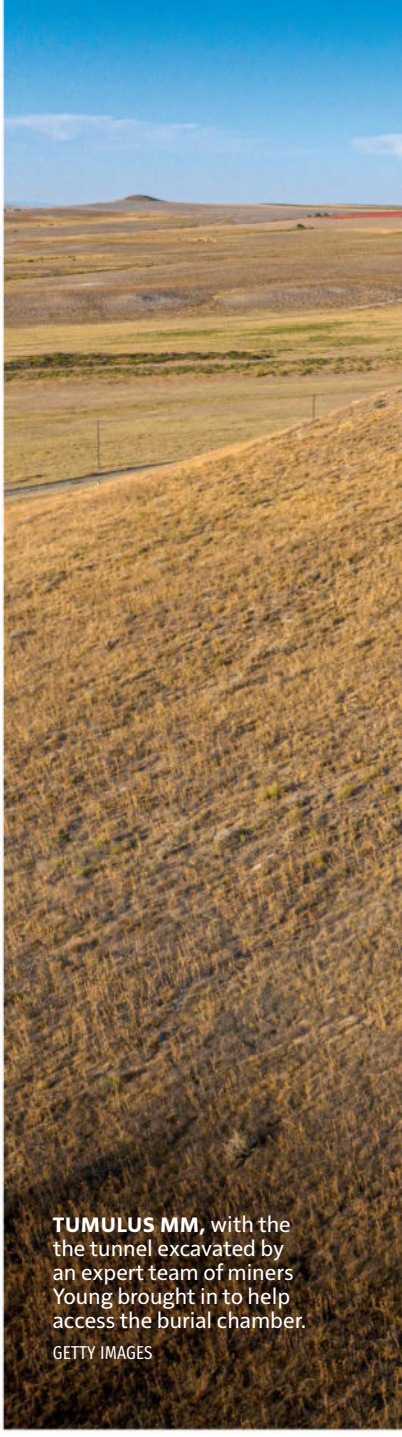
One of the mounds stands out from the rest: At nearly 175 feet high and a thousand feet in diameter, it dominates the landscape. This mound was traditionally thought to be the tomb of the most famous ruler of the Phrygian kingdom: Midas, who according to Greek myth turned all he touched into gold, including tragically his own daughter. Archaeologists dubbed the site Tumulus MM (for Midas Mound). As most barrows had been looted, it was believed that the largest one would have suffered the same fate. But in 1957 archaeologists were delighted to discover that this was not the case.

NGM MAPS

That year, a team of researchers from the University of Pennsylvania Museum of Archaeology and Anthropology, led by Rodney S. Young, began work on the giant mound. The excavation soon ran into a challenge of engineering: how to reach the center without causing a collapse. Young brought in a team of experienced miners from the region of Zonguldak, the only site in Türkiye with hard coal deposits. Digging a 230-foot-long tunnel, they managed to reach the center of the tumulus in just 25 days.

Inside the Tomb

There, Young and his team found that the Phrygians had dug a pit around 59 feet long and 53 feet wide. The pit's walls had been lined with large ashlars to protect the interior from moisture. Both the walls and the floor of the chamber had been packed with rubble made from a mixture of small stones, gravel, fine sand, and clay.



TUMULUS MM, with the the tunnel excavated by an expert team of miners Young brought in to help access the burial chamber.

GETTY IMAGES

Young's team carefully removed the ashlars and rubble, and as they did so, the sides of a wooden burial chamber appeared. This

A TUMULUS OVER TIME

8th century B.C.
A large burial mound is built in the necropolis of Gordion, near modern-day Ankara.

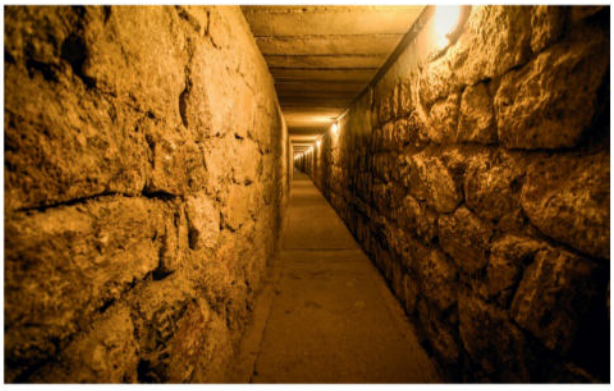
1957
Rodney Young begins excavating the Tumulus MM and finds an intact burial chamber inside.

1988
A new phase of excavation and research at Gordion begins after Young's death in 1974.

2023
UNESCO includes the archaeological site of Gordion on the World Heritage List.

consisted of a kind of airtight chest about 16 feet wide and almost 10 feet high, made of thick beams and planks of cedar, pine, and juniper. Since the Iron Age Phrygian engineers had sealed the tomb with a roof, and there was no other access, Young decided to cut an aperture in one of the wooden walls with enough room to insert one's head, an arm, and a flashlight in order to peer into the space, which had remained untouched for 2,700 years.

Once their eyes had adjusted to the darkness inside the wooden chamber, they noticed the characteristic green color of oxidized bronze everywhere. Despite King Midas's proverbial association with gold, the mineral was conspicuously absent; most of the grave goods (cauldrons, jugs, and drinking bowls) were made of bronze.

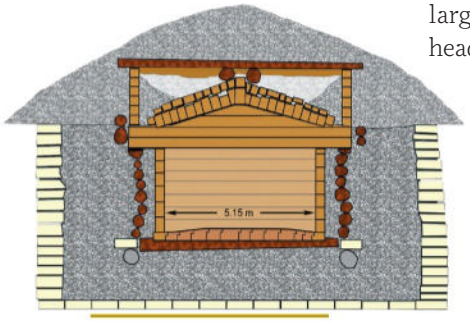


Inside the tumulus, a view down the tunnel excavated by Young's team.
BAYRAM ONUR BALLI/GETTY IMAGES

ENTRANCE to the burial chamber of Tumulus MM, cut by Rodney Young's team; from here they got their first glimpse inside.
ALAMY/CORDON PRESS

As it turned out, it wasn't the metal objects that most captivated the archaeologists but the well-preserved and intricately crafted wooden furniture placed inside the chamber. They found two wooden screens, or stands, with open-worked panels, two low stools, a chair on which fabrics had been placed, and nine three-legged tables.

In the northwest corner of the chamber, they found the remains of the deceased. Later analysis would reveal them to be from a man between 60 and 65 years old and of short stature: just over five feet tall. The body had been placed on a pile of dyed textiles inside a large cedar coffin, with the head facing east. The identity of the deceased is still to be confirmed, but radiocarbon dating suggests that he died around 740 B.C. This means he is unlikely to be the legendary King Midas, a figure scholars identify as being King Mita of Mushki. Based on documentation from his erstwhile enemy, Sargon II of Assyria, who must have died at the end of the eighth century B.C. Archaeologists now believe that the man is almost certainly Gordias, Midas's father and founder of Gordion. The tomb, therefore, might not belong to Midas, but was probably a work of his creation.

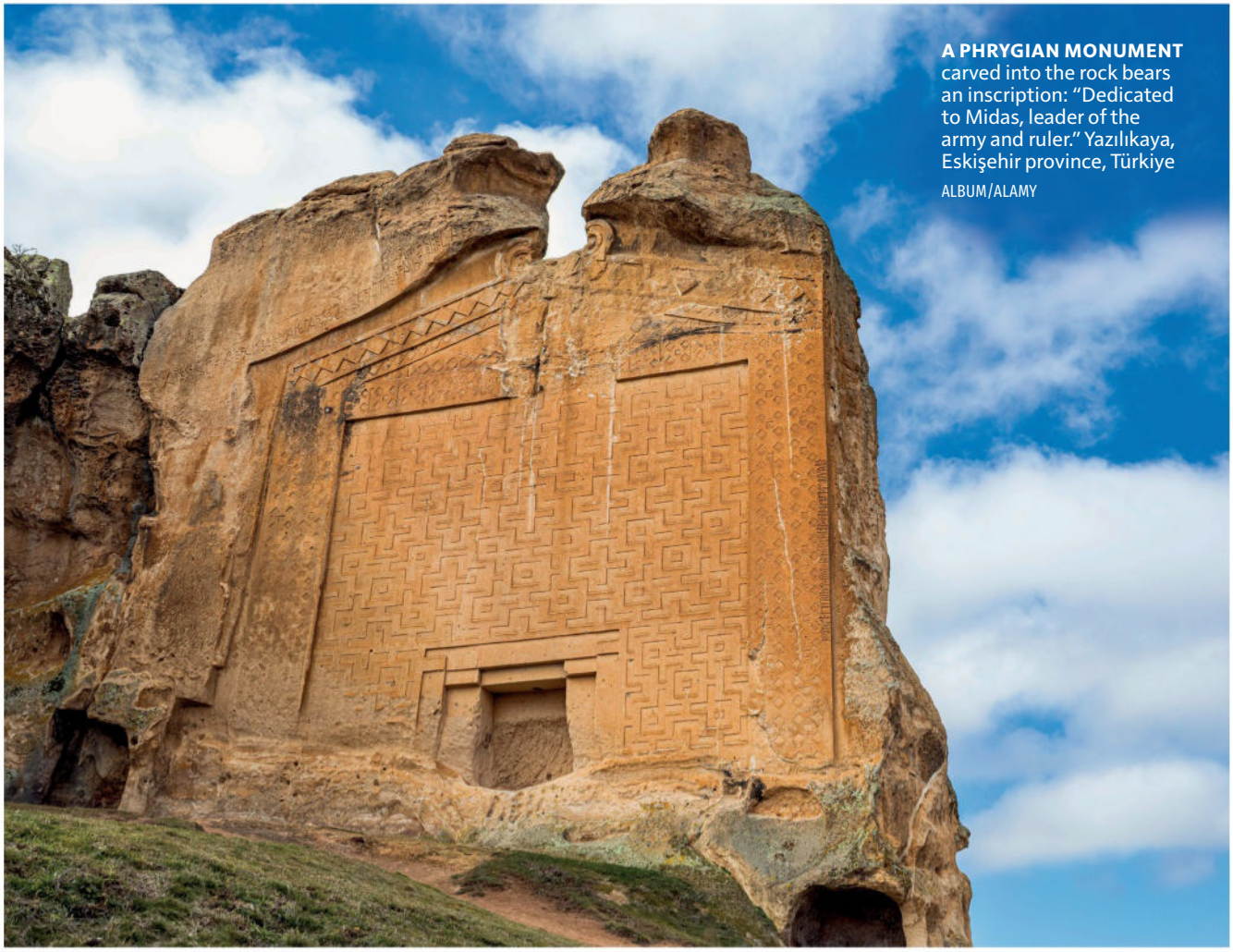


A Farewell Banquet

The grave goods in the Tumulus MM are extraordinary in terms of their quantity. In addition to two situlae (pail-like vessels), one in the shape of a lion's head and the other with the head of a ram, there are 10 bronze jugs (which, as evidence suggests, had been hung on the wall), 15 bowls with swivel handles, and a hundred other drinking

The skeleton of a man aged around 60 to 65 was discovered in a large cedar coffin.

Section of the burial chamber at Gordion with the cedar coffin.
ILLUSTRATION: RICHARD LIEBHART AND BANU BEDEL. COURTESY: PENN MUSEUM GORDION PROJECT



A PHRYGIAN MONUMENT carved into the rock bears an inscription: "Dedicated to Midas, leader of the army and ruler." Yazılıkaya, Eskişehir province, Türkiye

ALBUM/ALAMY

bowls. These figures may reflect the number of people who attended the funeral, each guest having their own bowl. Also found, tucked into a bag, were 145 bronze *fibulae* (garment pins) plus another 37 mixed in with the textiles inside the coffin. It has been suggested that these fibulae were farewell gifts left by the guests during the funeral. These items, as well as the tables and seats, were likely used in a funeral ceremony that involved a great banquet, which Midas himself would have presided over.

In the late 1990s, Patrick McGovern, a researcher at the University of Pennsylvania specializing in molecular archaeology, studied the organic residues from Tumulus MM and concluded that the guests at the funeral ate a stew of roasted sheep or goat and lentils, seasoned with various spices. To drink, they had a mix of wine, beer, and mead.

Once the deceased and his grave goods had been buried, the Phrygians covered the chamber with a gabled wooden structure, which was in turn covered with a wooden lintel. These carefully engineered additions were necessary to ensure that the tomb could withstand the enormous pressure of the clay, stone, and earth heaped over it. This contrivance is evidence of a monumental burial for a person of importance. Four names have been found inscribed on a roof beam in the burial chamber: Urunis, Nana, Myksos, and Sitsidos. The latter may well have been the architect of the tomb.

The Project Continues

After Rodney S. Young's tragic death in a traffic accident in 1974, the University of Pennsylvania's surveys at Gordion came to a halt. They later resumed and are ongoing: In 2025, another royal Phrygian tomb (T26) close to Tumulus MM was excavated. Over the years, excavations in and around Gordion have uncovered remains dating from the Early Iron Age to the Hellenistic period, and have revealed that the city was much larger than previously assumed. The importance of the site was recognized in 2023, when UNESCO included it on the World Heritage List.

—Ángel C. Aguaro Pérez

Gordion Lot

INSIDE THE MOUND, Young and his team found the remains of a magnificent burial hoard. Some of the objects are shown here.

1. ALAMY/CORDON PRESS; 2. ALBUM; 3. MET/SCALA, FLORENCE; 4 AND 5: O. LOUIS MAZZATENTA/NATIONAL GEOGRAPHIC IMAGE COLLECTION

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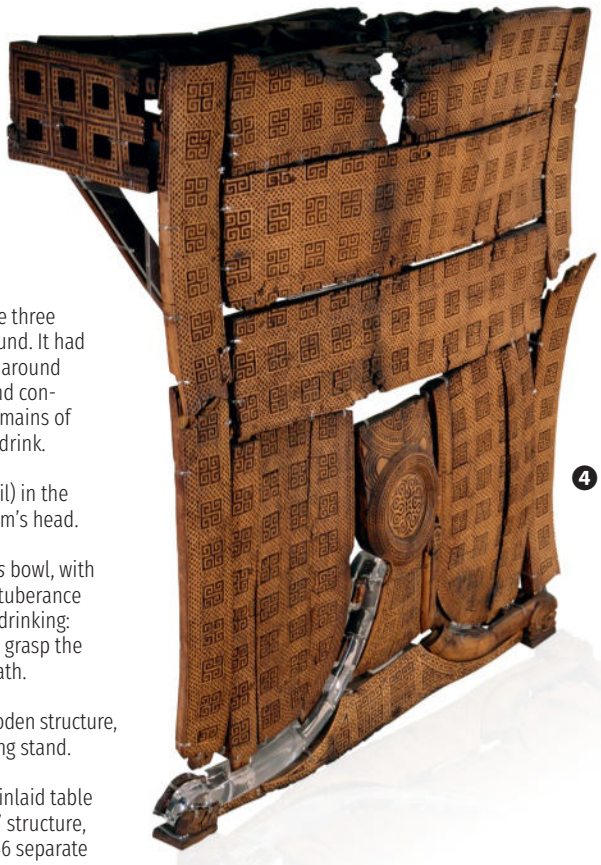

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❶ One of the three cauldrons found. It had a capacity of around 40 gallons and contained the remains of a fermented drink.

❷ *Situla* (pail) in the shape of a ram's head.

❸ *Omphalos* bowl, with a central protuberance which aided drinking: a hand could grasp the hollow beneath.

❹ Inlaid wooden structure, likely a serving stand.

❺ Restored inlaid table with 'Pagoda' structure, made up of 46 separate components.

❺

❸

The archaeological site Skara Brae (Orkney Islands, Scotland) holds the remains of a Neolithic village encompassing several small houses and what is thought to be a workshop. The discovery of artifacts and "furniture" construction offer a glimpse into the life of a Stone Age community.

MANUEL COHEN/AURIMAGES

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Sourced from the Baltic region—home to the world's largest deposits dating back 44 million years to the Eocene epoch—this remarkable gem captivates collectors with its prehistoric beauty. It is a genuine piece of natural history you can wear every day. Because natural amber develops organically, **each hand-set gem possesses its own unique golden depth**, ensuring you receive a truly one-of-a-kind original.

Cross the perfect gift off your holiday list today. Secure this pendant and matching complimentary chain for the **exclusive price of $99** (plus S&H). Want to truly surprise someone special? Choose the **complete Baltic Amber Feline Collection for only $159** (plus S&H). You instantly **save $40** while giving a complete luxury set ready for gifting.

Call toll-free or visit us online today and quote your promo code to claim this exclusive offer. Shipped directly to your door in a presentation box, it makes holiday gifting completely effortless. Order now to secure this genuine piece of 44-million-year-old natural history while supplies last!

'Great Christmas Gift'

"I gave the pendant to my wife for Christmas. She lost her beloved cat a few months ago and the pendant reminded her of him. Thank you."

- B. Crane, KY

★★★★★

Original Price Without Code ~~$249~~ - With Promo Code **Now Only $99**

A. Baltic Amber Feline Pendant and *FREE Sterling Silver Chain* - $99 plus S&H

B. Baltic Amber Feline Earrings - $99 plus S&H

Baltic Amber Feline Complete Collection - **Now Only $159** (SAVE AN EXTRA $40)

ORDER NOW TOLL FREE 24/7 ON: **1-800 733 8463**

OR ORDER ONLINE AT: **timepiecesusa.com/ng6nbc** AND ENTER CODE: **NG6NBC**

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